



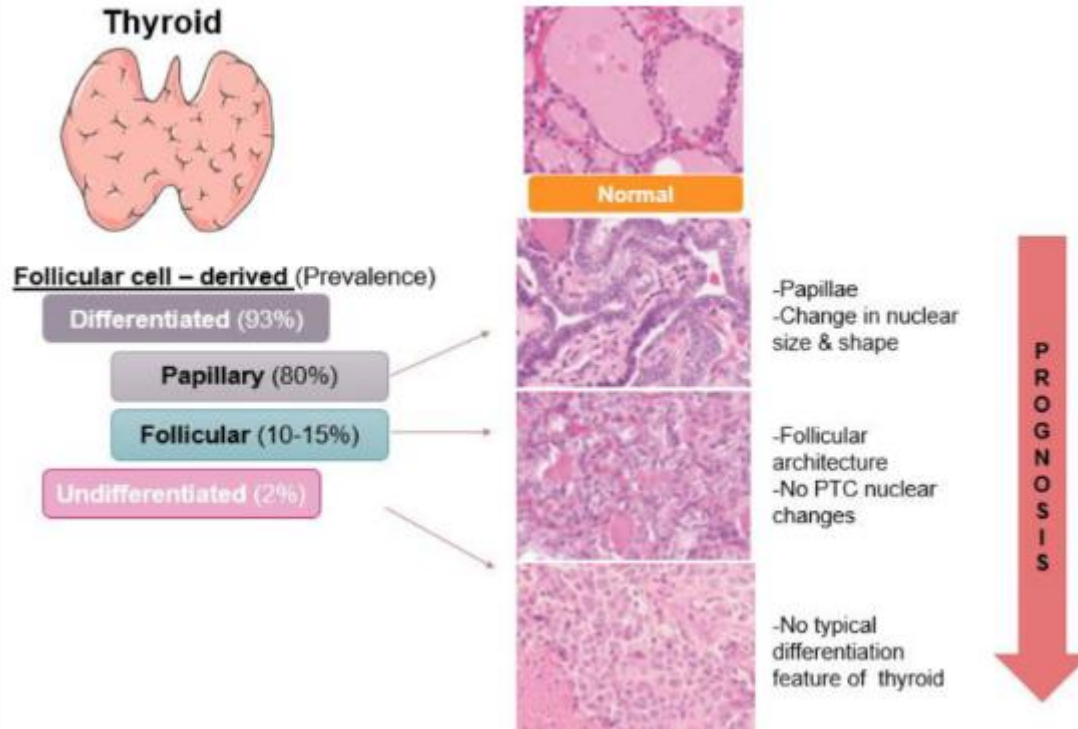
KUZMAN CONSULTING

BIOINFORMATICIANS AT YOUR SERVICE

Naši podatci: GSE224356

- Tao M, Ruan X, Yu J, Dong Q et al. Enhancer-mediated NR2F2 recruitment activates BGN to promote tumor growth and shape tumor microenvironment in papillary thyroid cancer. *Theranostics* 2026;16(1):298-324. PMID: [41328346](#)
- single-end RNA-seq papilarnog karcinoma štitnjače

Papilarni karcinom štitnjače (PTC)



Naši podatci: GSE224356

- RNA-seq papilarnog karcinoma štitnjače
- upareno sa H3K27ac ChIP-seq-om → *multi-omics approach*
- povećana ekspresija BGN u malignim stanicama povezana s lošijom kliničkom slikom; BGN knock-down inducira arest staničnog ciklusa
- NR2F2 regulira ekspresiju BGN, a BGN aktivira signalni put AKT → rast tumora

Nuclear receptor subfamily 2 group F member 2 (NR2F2)

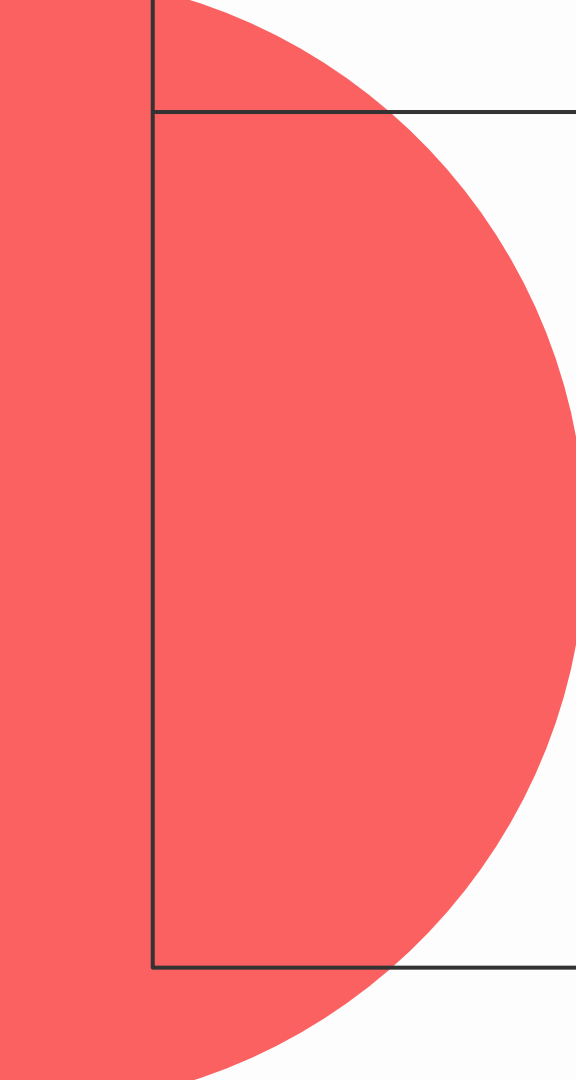
- <https://www.ncbi.nlm.nih.gov/gene/7026>
- nuklearni receptor steroidnih hormona štitnjače
- TF induciran ligandom

Cited Variations, dbSNP b157 v2		rs3743462 A/G/T		rs58777722 A/C/G/T		rs58777721 C/A		rs32845 G/A/C/T	
Live RefSNPs, dbSNP b157 v2		rs3743462 A/G/T		rs58777722 A/C/G/T		rs58777721 C/A		rs32845 G/A/C/T	
ClinVar variants with precise endpoints		rs3743462 A/G/T		rs58777722 A/C/G/T		rs58777721 C/A		rs32845 G/A/C/T	
3347446 A	1262707 T	918834 -	3095670 -	617884 -					
4407291 G/T		988261 -	3799936 A	1261378 T					
426701 A		1240299 T	4487299 C/G	5286443 -					
1291706 T		958490 T	1597254 T	1228987 ATTC					
		3768440 A	3748402 C	4023242 G					
		1066990 G	3088856 T	1547906 T					
		2170640 A	4626309 T	91206 C					
		1177624 A	6086266 T	73561 T					
		4122406 A	3996685 C/G	338447 G					
		3817101 T	3853568 A	2429791 -					
		521441 T	1338920 T	128252 A					
		2259072 C	391115 C	3719233 T					
		5471081 G/C	4371845 T	3873584 T					
		475785 A	1528649 A	182281 C					
		2042346 G	1305520 T/G	625980 A					
		2541489 T	3778737 T	3361736 T					
		3487657 A	3862062 -	2442212 A					
		3928482 G	3788452 A	521371 C					
		3252829 T	2881443 T	283321 G					
		241367 A	4886768 T	4487482 C/A					
		306891 A	1452067 T	461238 T					
		4722378 GAGCCCGCCGCGCCG - 52258 C		3361791 T					
		247528 A	128211 T	2578647 GAT					
		2967281 T	273396 C	4750172 A					
		2357274 C	2862479 A	3862234 AT					
		907358 C	235558 A	4122407 C					
		3487659 C	3487660 A	2583816 G					
		2834141 T	3381338 T	1555684 T					
		2561942 G	146751 A	2862452 T					
		2113678 A	4751787 G	1381821 G					
		648947 G	3838480 T	1286927 -					
		125254 AGGC	3862523 -						
		166935 A	2048295 T						
		948591 T	1321294 A						
		937696 T	1321348 T						
		3038853 T	3888912 A						
		448122 T	242178 T						
		4814288 T	2882753 A						
		3733829 A	4749812 T						
		1537868 G	598763 A						
		1864859 G	3668258 C						
		1723583 C	2276848 T						
		2578648 T	3662896 G						
		854882 T	1434821 T						
		2578643 A	4754843 C						
		2662817 G	679985 G						
		3359688 T	2054020 T						
		3897277 G	3888927 T						
		1014352 G	413740 A						
		9176642 A	1476770 A						

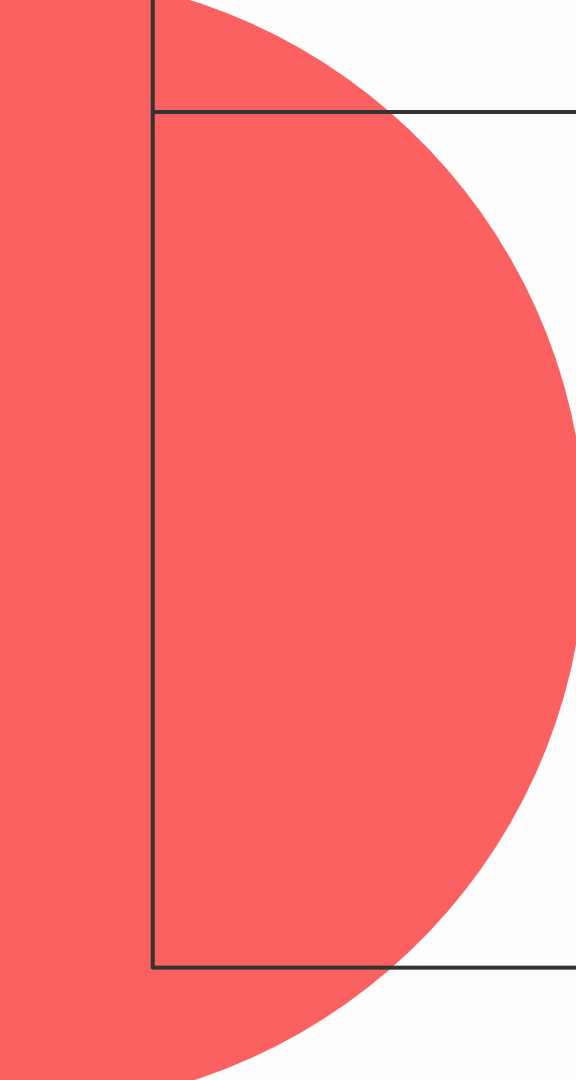
Biglikan (BGN)

- <https://www.ncbi.nlm.nih.gov/gene/633>
- SLRP (small leucine-rich proteoglycan)
- uloga u rastu kostiju, razvoju i regeneraciji mišića, i organizaciji kolagenskih vlakana
- ECM



A large red circular graphic element is positioned on the left side of the slide, partially overlapping the white rectangular area. It is a solid red color and appears to be a semi-circle or a large arc.

Rad na platformi Galaxy

A large red circle is partially visible on the left side of the slide, overlapping the white content area.

0. Učitavanje podataka u Galaxy

NCBI

Resources

How To

Sign in to NCBI

GEO Home

Documentation

Query & Browse

Email GEO

Gene Expression Omnibus

GEO is a public functional genomics data repository supporting MIAME-compliant data submissions. Array- and sequence-based data are accepted. Tools are provided to help users query and download experiments and curated gene expression profiles.

Getting Started

- Overview
- FAQ
- About GEO DataSets
- About GEO Profiles
- About GEO2R Analysis
- How to Construct a Query
- How to Download Data

Tools

- Search for Studies at GEO DataSets
- Search for Gene Expression at GEO Profiles
- Search GEO Documentation
- Analyze a Study with GEO2R
- Studies with Genome Data Viewer Tracks
- Programmatic Access
- FTP Site
- ENCODE Data Listings and Tracks

Browse Content

Repository Browser

DataSets:	4348
Series:	281794
Platforms:	28268
Samples:	8447411

Information for Submitters

- Login to Submit

- Submission Guidelines
- Update Guidelines

- MIAME Standards
- Citing and Linking to GEO
- Guidelines for Reviewers
- GEO Publications

GSE224356

NCBI GEO

GSE224356


Gene Expression Omnibus

HOME | SEARCH | SITE MAP | GEO Publications | FAQ | MIAME | Email GEO

NCBI > GEO > [Accession Display](#) [?] Not logged in | [Login](#) [?]

Scope: Format: Amount: GEO accession:

Series **GSE224356** [Query DataSets for GSE224356](#)

Status	Public on May 27, 2025
Title	RNA-seq of papillary thyroid carcinoma (PTC) patient's tissues and paired normal tissues
Organism	Homo sapiens
Experiment type	Expression profiling by high throughput sequencing
Summary	To describe the transcriptome of PTC, we performed RNAseq analysis of PTC patient's tissues and paired normal tissues
Overall design	RNA-seq
Contributor(s)	Ruan X , Zhang W , Hou X
Citation(s)	Tao M, Ruan X, Yu J, Dong Q et al. Enhancer-mediated NR2F2 recruitment activates BGN to promote tumor growth and shape tumor microenvironment in papillary thyroid cancer. <i>Theranostics</i> 2026;16(1):298-324. PMID: 41328346
Submission date	Feb 02, 2023
Last update date	Jan 27, 2026
Contact name	Xiukun Hou
E-mail(s)	xiukunhou@gmail.com
Organization name	Tianjin Cancer Institute and Hospital
Street address	Huanhu Road
City	Tianjin
ZIP/Postal code	30000
Country	China

Platforms (1)

[GPL20795](#) HiSeq X Ten (Homo sapiens)

Samples (6)

[GSM7020412](#) Normal tissue_1

[GSM7020413](#) Tumor tissue_1

[GSM7020414](#) Normal tissue_2

[# More...](#)

This SubSeries is part of SuperSeries:

[GSE224357](#) H3K27ac ChIP-seq and RNA-seq of papillary thyroid carcinoma patient's tissues and paired normal tissues

Relations

BioProject [PRJNA930711](#)

Download family

[SOFT formatted family file\(s\)](#)

[MINiML formatted family file\(s\)](#)

[Series Matrix File\(s\)](#)

Format

SOFT [?](#)

MINiML [?](#)

TXT [?](#)

Supplementary file	Size	Download	File type/resource
GSE224356_T1_N1_Down_genes.xlsx	1.9 Mb	(ftp) (http)	XLSX
GSE224356_T1_N1_Up_genes.xlsx	1.3 Mb	(ftp) (http)	XLSX
GSE224356_T2_N2_Down_genes.xlsx	849.5 Kb	(ftp) (http)	XLSX
GSE224356_T2_N2_Up_genes.xlsx	1.2 Mb	(ftp) (http)	XLSX
GSE224356_T3_N3_Down_genes.xlsx	892.5 Kb	(ftp) (http)	XLSX
GSE224356_T3_N3_Up_genes.xlsx	1.2 Mb	(ftp) (http)	XLSX

[SRA Run Selector](#) [?](#)

Raw data are available in SRA

Processed data are available on Series record

SRA Run Selector



Accession

PRJNA930711



Search

Common Fields

BioProject	PRJNA930711
Consent	PUBLIC
Assay Type	RNA-Seq
AvgSpotLen	50
Center Name	TIANJIN CANCER INSTITUTE AND HOSPITAL
DATASTORE filetype	FASTQ, RUN.ZQ, SRA
DATASTORE provider	GS, NCBI, S3
DATASTORE region	gs.us-east1, ncbi.public, s3.us-east-1
Instrument	HiSeq X Ten

Select

	Runs	Bytes	Bases	Download	Cloud Data Delivery	Computing
Total	6	5.10 Gb	6.50 G	Metadata or Accession List		
Selected	0	0	0	Metadata or Accession List or JWT Cart	Deliver Data	Galaxy

Galaxy Europe

Using 2% of 250.0 GB | dunjag

Upload from Disk or Web to **Unnamed history**

Regular Composite Collection Rule-based

Drop files here

Type (set all): Auto-detect Reference (set all): unspecified (?)

Choose local file Choose from repository Paste/Fetch data Start Pause Reset Close

« < 1 > » 10 per page, 10 total

History

search datasets

Unnamed history

Default Storage

0 B

This history is empty.
You can load your own data or get data from an external source.

vaši podatci

Galaxy Europe

Upload

Tools

ChatGXY

Workflows

Workflow Invocations

Interactive Tools

Visualization

Histories

History Multiview

Datasets

Pages

Tools

Discover Tools

search tools

Get Data

Send Data

Collection Operations

GENERAL TEXT TOOLS

Text Manipulation

Convert Formats

Filter and Sort

Join, Subtract and Group

GENOMIC FILE MANIPULATION

Convert Formats

FASTA/FASTQ

FASTQ Quality Control

Quality Control

SAM/BAM

BED

VCF/BCF

Nanopore

COMMON GENOMICS TOOLS

Using 10% of 250.0 GB

dunjag

Upload from Disk or Web to Radionica

RegularCompositeCollectionRule-based

SRR_Acc_List.txt72 bAuto-detectorspecified (?)100%

Type (set all):Auto-detectorspecified (?)Reference (set all):unspecified (?)

Choose local file

Choose from repository

Paste/Fetch data

Start

Pause

Reset

Close

News

More news

Events

More events

Galaxy 26.0 is here!

Galaxy 26.0 brings major new capabilities to the Galaxy experience, from built-in AI assistance and interactive visualization to smarter workflow completion and easier troubleshooting. Explore the user release notes.

BioNT Brings Galaxy Training to German, Spanish, and Italian Learners

We are thrilled to announce that the BioNT project (Bio Network for Training) has contributed a significant set of human-curated translated Galaxy training materials to the GTN, now live and

Small Scale Galaxy Admins Meeting

Apr 16 · Online, Global

19th OME Community Meeting 2026

Apr 28 · Haus der Universität, Düsseldorf, Germany

Join us at the 19th OME Community Meeting for a workshop on image analysis in Galaxy and the OMERO integration.

Upload

Tools

ChatGXY

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Nanopore

COMMON GENOMICS TOOLS

Operate on Genomic Intervals

Fetch Sequences / Alignments

GENOMICS ANALYSIS

Annotation

Galaxy France

Galaxy Australia

Galaxy Europe

Galaxy

Galaxy has no borders

News

More news

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Galaxy Metagenome Training Course 2026

Outcomes of the 2026 Metagenome Training Course

Events

More events

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European Geosciences Union (EGU) 2026

May 4 · Vienna

A Data Terra session to the European Geoscience Union

History

search datasets

Unnamed history

Default Storage

6 B

1: SRR_Acc_List.txt

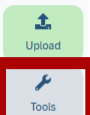
siva kućica = u redu čekanja

narančasto = proces se izvršava

zeleno = proces je uspješno završio

crveno = došlo je do greške

The screenshot displays the Galaxy Training Network interface. The top navigation bar is blue with the 'Galaxy Europe' logo on the left and user information 'dunjag' on the right. The main content area features a large orange star logo with the text 'Learn Galaxy with Galaxy Training Network'. On the left, a sidebar contains navigation links: Upload, Tools, ChatGXY, Workflows, Workflow Invocations, and Interactive. The 'Tools' section is expanded, showing a search bar and a list of tool categories: Get Data, Send Data, Collection Operations, GENERAL TEXT TOOLS (highlighted), Text Manipulation, Convert Formats, Filter and Sort, and Join, Subtract and Group. On the right, a 'History' panel shows a list of datasets. The dataset '1: SRR_Acc_List.txt' is highlighted with a red box, indicating it is the current selection. The status bar at the bottom indicates 'Using 10% of 250.0 GB'.



Tools

Discover Tools

search tools

Get Data

Adds environment variables From Copernicus and etopo given geolocalized and timestamped observations

AquaINFRA Importer downloads content via the AquaINFRA Interaction platform

Argo data access for global ocean in situ observing system

BioMart Ensembl server

BioMart Test server

CBI Rice Mart rice mart

Copernicus Marine Data Store retrieve marine data

DICED Database DICED is a web interface for access to proteolytic peptides

DoRiNA Search DoRiNA data source

Download and Extract Reads in BAM format from NCBI SRA

Download and Extract Reads in FASTQ format from NCBI SRA

Download and Generate Fileup Format from NCBI SRA

Download run data from EBI Metagenomics database

downloads via lftp

EBI Search to obtain search results on

Download and Extract Reads in FASTQ format from NCBI SRA (Galaxy Version 3.1.1+galaxy1)

Tool Parameters

select input type

List of SRA accession, one per line

sra accession list *

1: SRR_Acc_List.txt

accepted formats

Select output format *

- ☒ gzip compressed fastq
☐ Uncompressed fastq
☐ bzip2 compressed fastq

Compression will greatly reduce the amount of space occupied by downloaded data. Downstream applications such as a short-read mappers will accept compressed data as input. Consider this example: an uncompressed 400 Mb fastq datasets compresses to 100 Mb or 80 Mb by gzip or bzip2, respectively. (--gzip --bzip2)

Advanced Options

Additional Options

Email notification

☐ No

Send an email notification when the job completes.

Attempt to re-use jobs with identical parameters?

☐ No

This may skip executing jobs that you have already run.

Output Tags

Add tags

Enter tags to apply to the output datasets in your history (e.g., 'sample1'). Tags starting with '#' (e.g., '#sample1') will propagate to datasets derived from these outputs. Tags help you organize and search your history.

Run Tool

History

search datasets

Unnamed history

Default Storage

72 B

1: SRR_Acc_List.txt

Tools

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GENERAL TEXT TOOLS

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FASTQ Quality Control

Quality Control

SAM/BAM

BED

VCF/BCF

Nanopore



"Anyone, anywhere in the world should have free, unhindered access to not just my research, but to the research of every great and enquiring mind across the spectrum of human understanding." – Prof. Stephen Hawking

News

[More news](#)

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[More events](#)

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Apr 29 · Univ. der Universität, Düsseldorf, Germany

History

search datasets

RNAseq

Default Storage

5.86 GB

3: Single-end data (fastq-du mp)

a list with 6 fastqsanger.gz datasets

2: Paired-end data (fastq-du mp)

a list with 0 pairs

1: SRR_Acc_List.txt

Single-end data (fastq-dump)



a list with 6 **fastqsanger.gz** datasets



Download



Show
Details



Run Job
Again

1: **SRR23318468**



2: **SRR23318469**



3: **SRR23318470**



4: **SRR23318471**



5: **SRR23318472**



6: **SRR23318473**



Paired-end data (fastq-dump)



a list with 0 pairs



Download



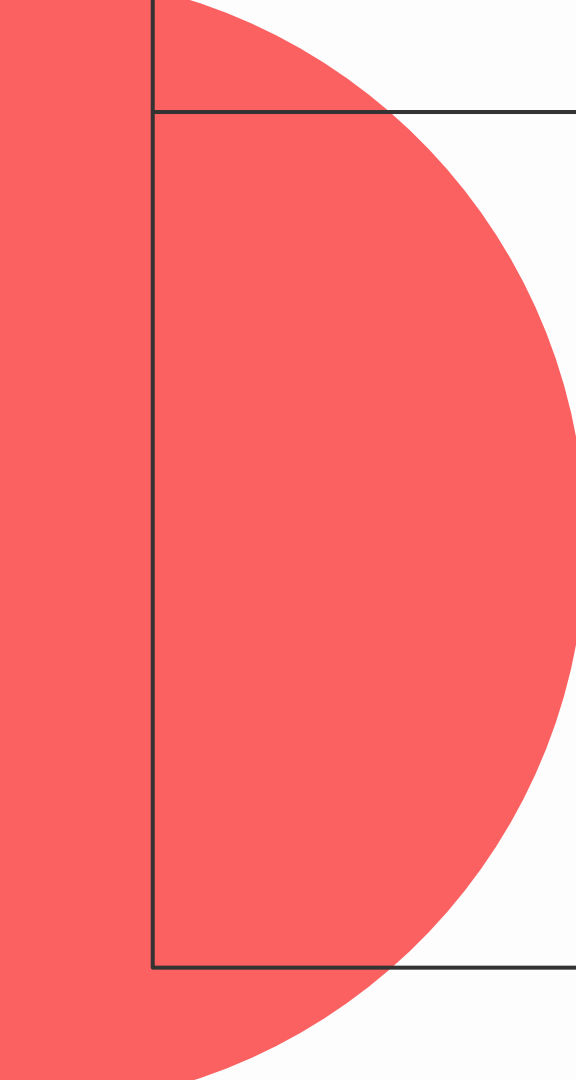
Show
Details



Run Job
Again

This is an empty collection.

samo single-end data pa je
paired-end empty collection

A large red circle is positioned on the left side of the slide, partially cut off by the edge. It overlaps with a white rectangular area that contains the text.

Genska ekspresija

Genska ekspresija

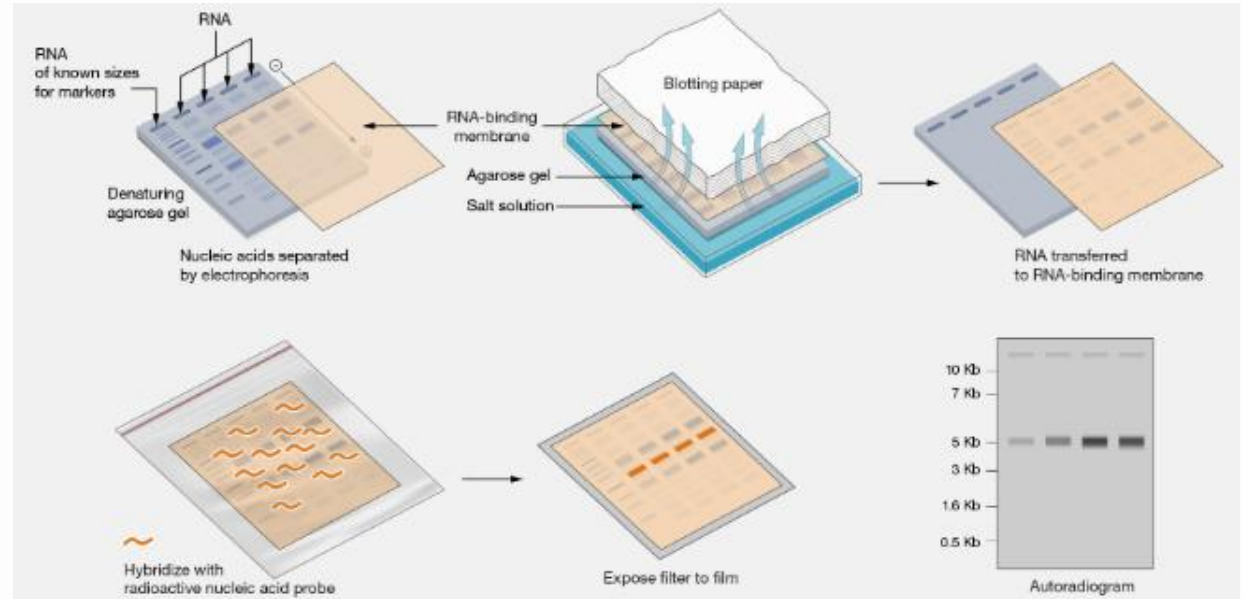
- proces kojim se informacija kodirana u genu pretvara u funkcionalni produkt (RNA ili protein)
- određuje koji se biokemijski procesi odvijaju u stanici
- odnos količine mRNA i produkta nije nužno 1:1, ali dovoljno dobra aproksimacija u većini slučajeva



praćenje ekspresije preko prisutnosti i količine mRNA

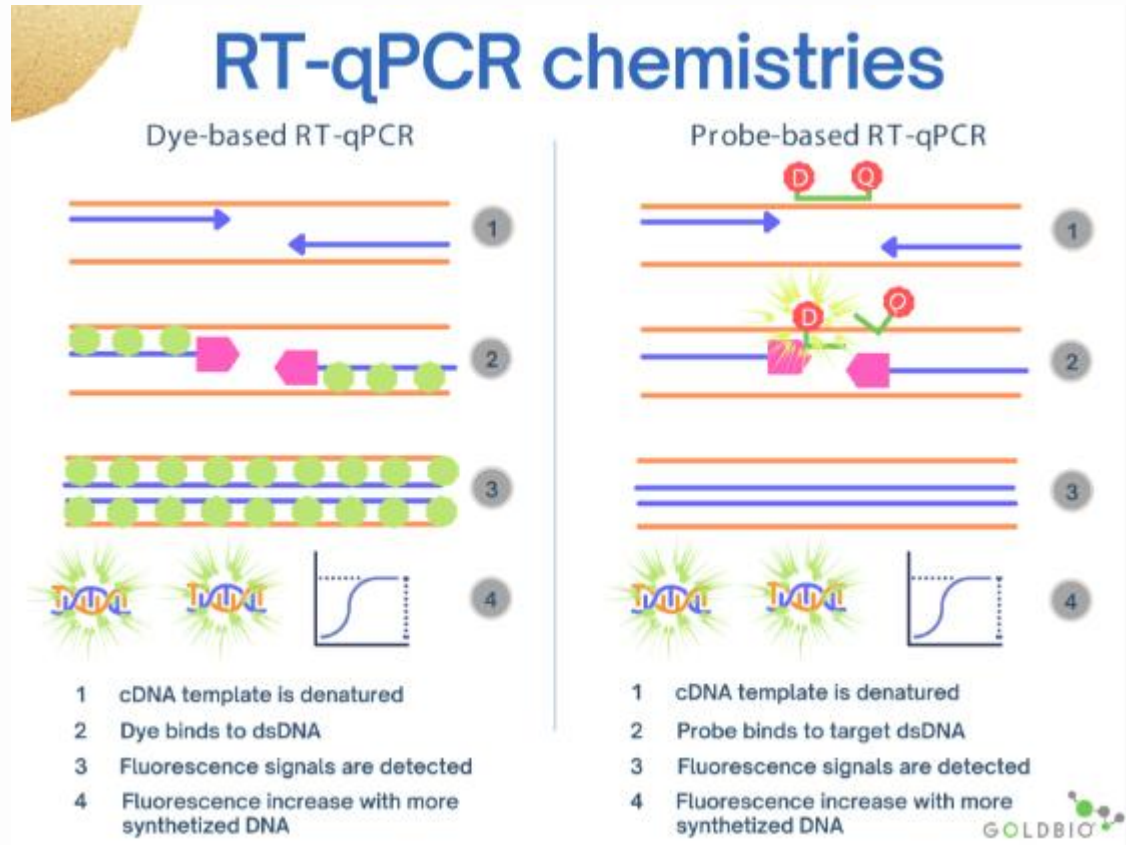
Metode

- northern blot
- RT-qPCR
- microarray
- SAGE



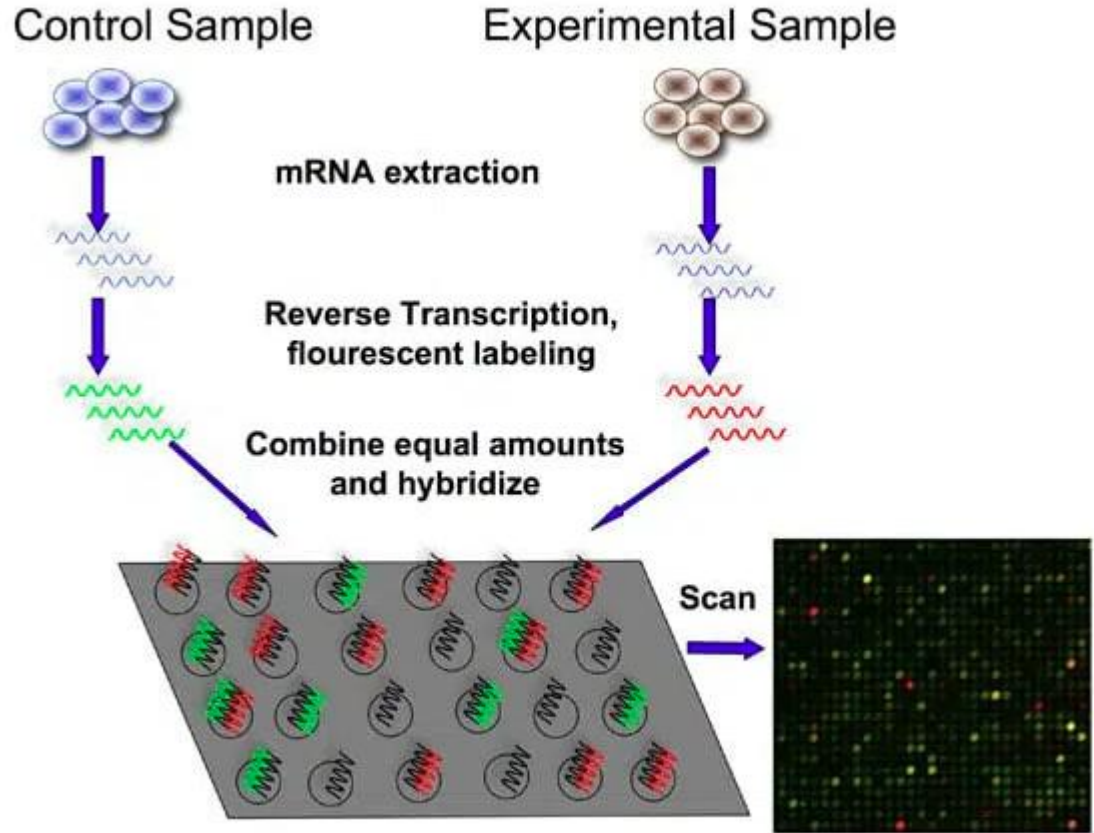
Metode

- northern blot
- **RT-qPCR**
- microarray
- SAGE



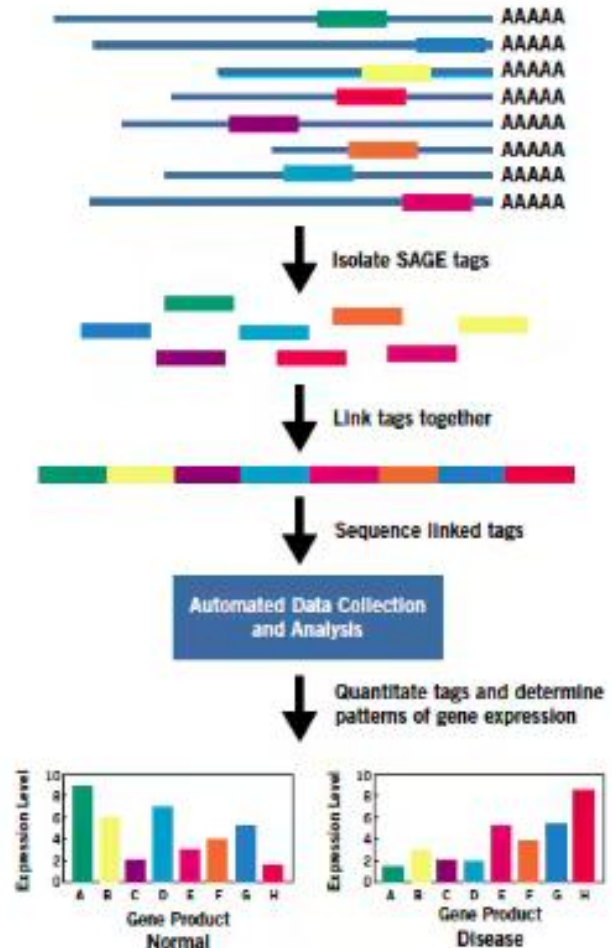
Metode

- northern blot
- RT-qPCR
- microarray
- SAGE



Metode

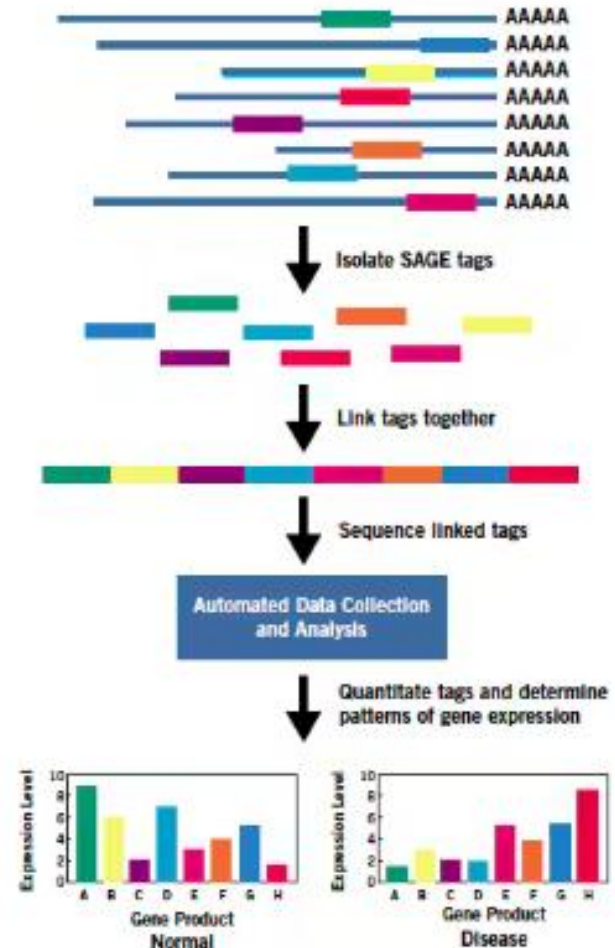
- northern blot
- RT-qPCR
- microarray
- **SAGE**



Metode

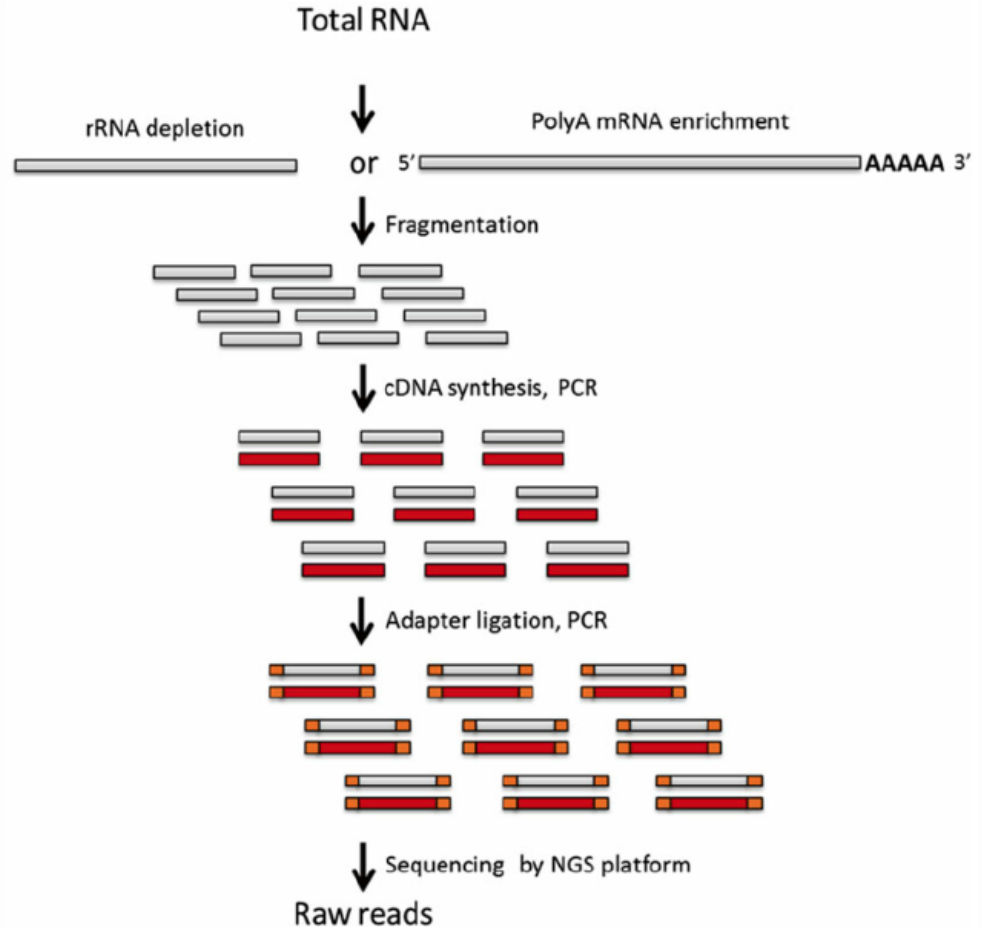
- northern blot
- RT-qPCR
- microarray
- SAGE

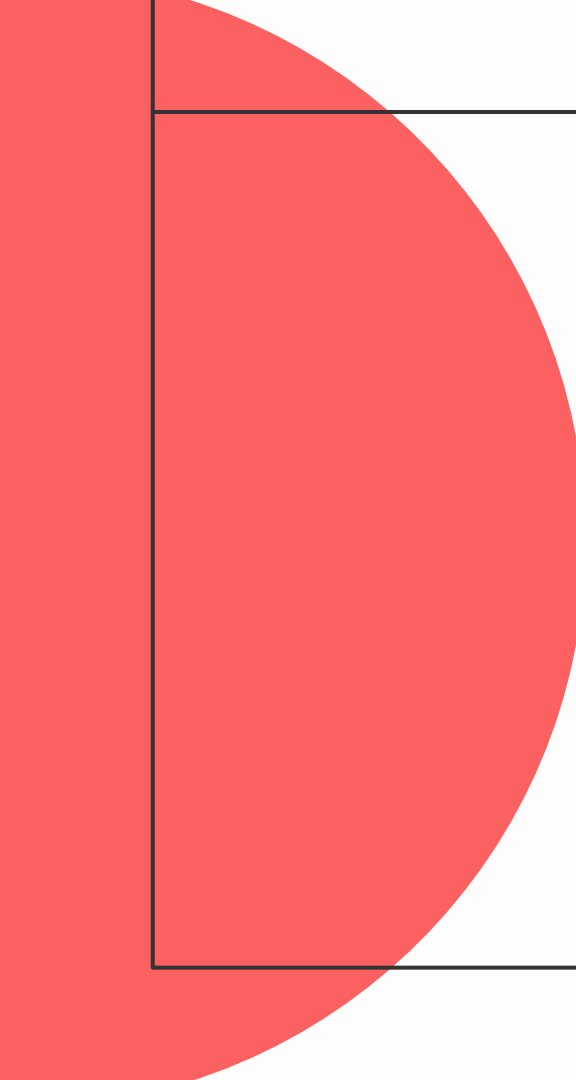
Minus? Unprijed određene sekvence koje možemo pronaći, i/ili nemogućnost otkivanja varijanti.



RNA-Seq

- nespecifične probe
 - “hvataju” sve
 - niža cijena od specifičnih proba
- brojne varijante
- moderni standard



A large red circle is partially visible on the left side of the slide, overlapping the white rectangular area.

Bioinformatička analiza genske ekspresije



Pitanja

1. Koje transkripte imamo?
2. U kojoj količini?

Pitanja

1. Koje transkripte imamo?

 mapiranje na genom

2. U kojoj količini?

 kvantitativna analiza

Pitanja

1. Koje transkripte imamo?



mapiranje na genom

= proces kojim određujemo
lokacije i relativne udaljenosti

2. U kojoj količini?

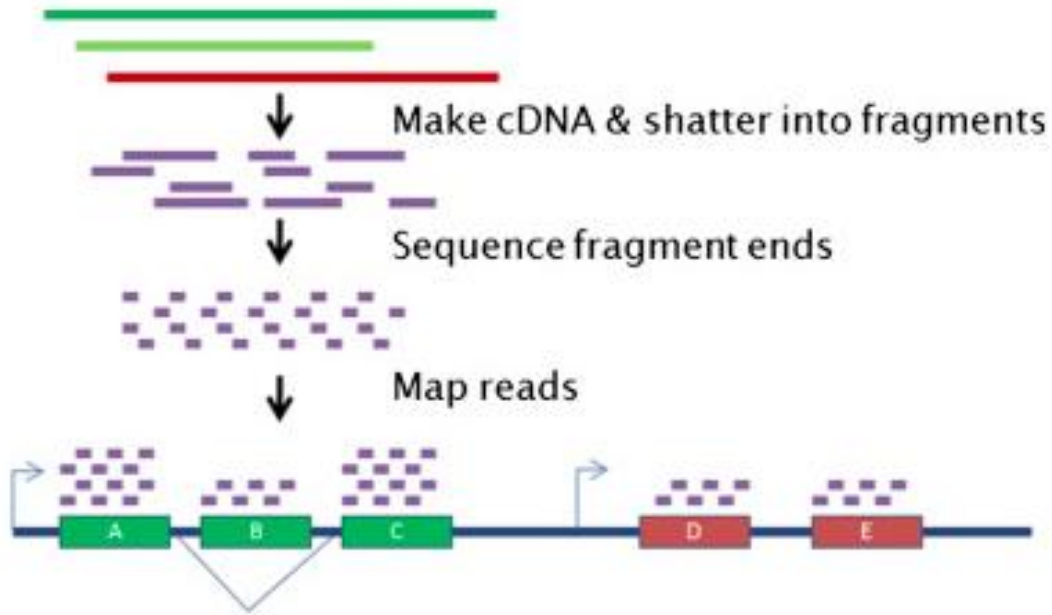


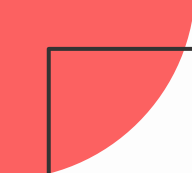
kvantitativna analiza

očitanih sekvenci na
kromosomima

Osnovni koraci

1. mapiranje na genom
2. featureCounts





Krećemo? NE!



=





Osnovni koraci

1. QC – kontrola sirovih podataka
2. mapiranje na genom
3. feature counts

Osnovni koraci

1. QC – kontrola sirovih podataka
2. kraćenje i pročišćavanje očitavanja (*trimming and filtering*)
3. mapiranje na genom
4. feature counts

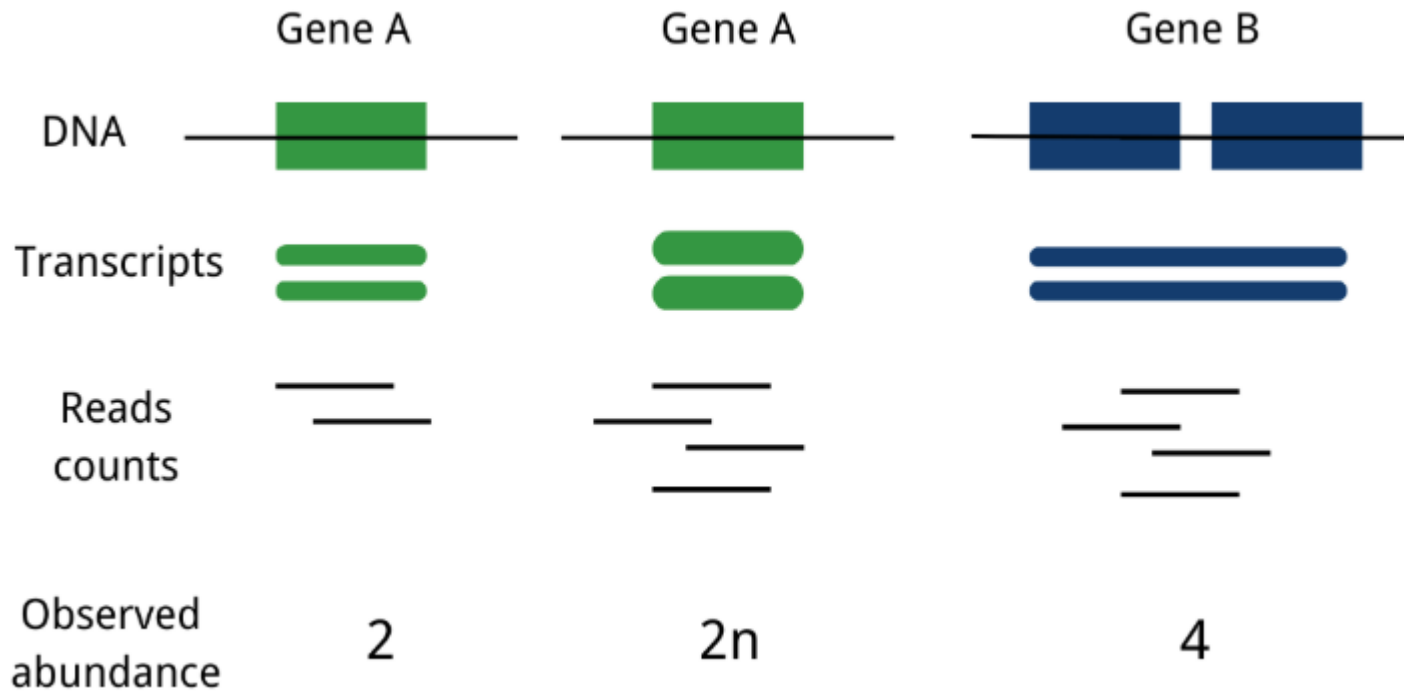
Osnovni koraci

1. QC – kontrola sirovih podataka
2. kraćenje i pročišćavanje očitavanja (*trimming and filtering*)
3. ponovna provjera kvalitete
4. mapiranje na genom
5. feature counts

Osnovni koraci

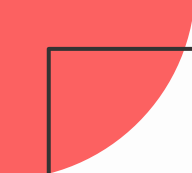
1. QC – kontrola sirovih podataka
2. kraćenje i pročišćavanje očitavanja (*trimming and filtering*)
3. ponovna provjera kvalitete
4. mapiranje na genom
5. alignment QC – kontrola mapiranja
6. feature counts

Osnovni koraci

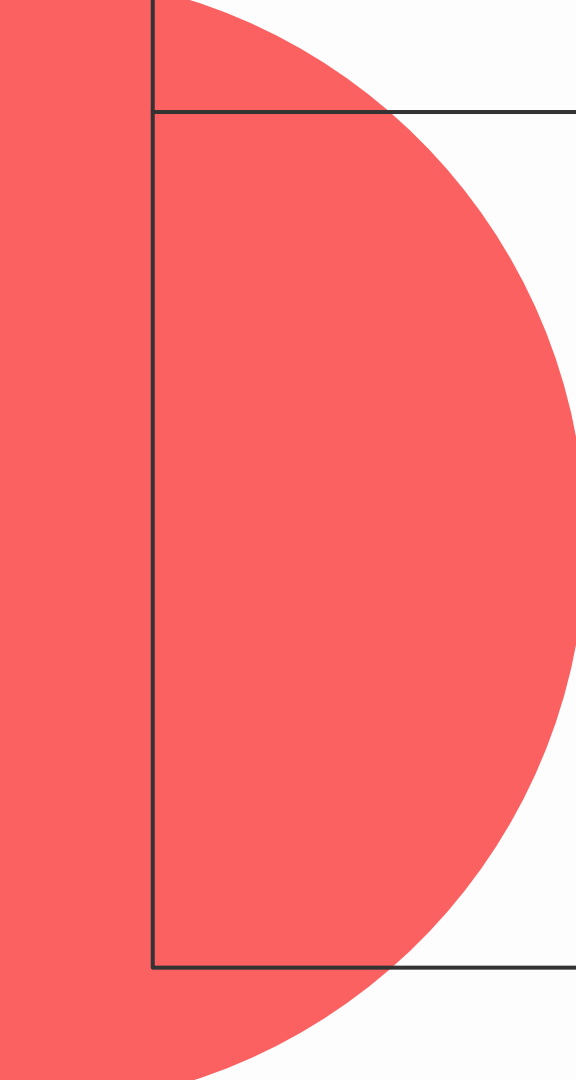


Osnovni koraci

1. QC – kontrola sirovih podataka
2. kraćenje i pročišćavanje očitavanja (*trimming and filtering*)
3. ponovna provjera kvalitete
4. mapiranje na genom
5. alignment QC – kontrola mapiranja
6. feature counts
7. normalizacija i statistička analiza



Krećemo!

A large red shape on the left side of the slide, consisting of a vertical rectangle and a semi-circle to its right.

1. Kontrola kvalitete podataka

Kontrola kvalitete sirovih *readova*: FastQC

Galaxy Europe

Using 10% of 250.0 GB dunjag

- Upload
- Tools
- ChatGXY
- Workflows
- Workflow Invocations
- Interactive Tools
- Visualization
- Histories
- History Multiview
- Datasets
- Pages

Tools

Discover Tools

fastqc

Show Sections

FastQC Read Quality reports **use Falco instead**

Falco An alternative, more performant implementation of FastQC for high throughput sequence quality control

Tabular to FASTQ converter

Make.fastq Convert fasta and quality to fastq

Tabular to FASTQ converter

FastQC Read Quality reports (Galaxy Version 0.74+galaxy1)

We recommend using **Falco** instead. Falco is 3 times faster and offers the same functionality as FASTQC.

Tool Parameters

! Please provide a value for this option.

Raw read data from your current history **required**

accepted formats

No compatible datasets available

Contaminant list - optional

Nothing selected

tab delimited file with 2 columns: name and sequence. For example: Illumina Small RNA RT Primer CAAGCAGAAGACGGCATACGA

Adapter list - optional

Nothing selected

List of adapters adapter sequences which will be explicitly searched against the library. It should be a tab-delimited file with 2 columns: name and sequence. (--adapters)

Submodule and Limit specifying file - optional

Nothing selected

a file that specifies which submodules are to be executed (default=all) and also specifies the thresholds for the each submodules warning parameter

Disable grouping of bases for reads >50bp

No

Using this option will cause fastqc to crash and burn if you use it on really long reads, and your plots may end up a ridiculous size. You have been warned! (--nogroup)

History

search datasets

RNAseq

Default Storage

5.86 GB

3 6

3: Single-end data (fastq-du mp)

a list with 6 fastqsanger.gz datasets

2: Paired-end data (fastq-du mp)

a list with 0 pairs

1: SRR_Acc_List.txt

Kontrola kvalitete sirovih *readova*: FastQC

Tool Parameters

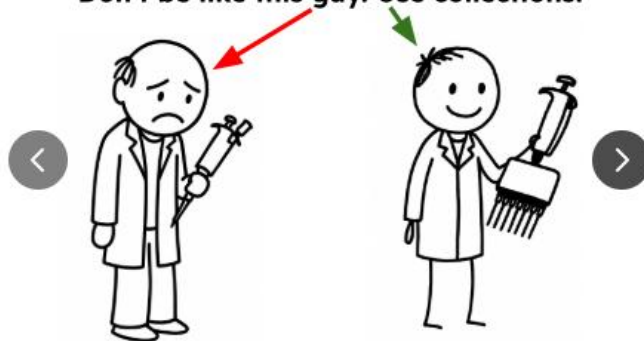
Raw read data from your current history *

    3: Single-end data (fastq-dump) 

accepted formats ▼

! The supplied input will be mapped over this tool.

Don't be like this guy. Use collections!



Kontrola kvalitete sirovih *readova*: FastQC

FastQC Read Quality reports (Galaxy Version 0.74+galaxy1) ☆ 🔗 ▶ Run Tool

Contaminant list - optional

📄 📄 📁 ⋮ Nothing selected

accepted formats ▼

tab delimited file with 2 columns: name and sequence. For example: Illumina Small RNA RT Primer CAAGCAGAAGACGGCATACGA

Adapter list - optional

📄 📄 📁 ⋮ Nothing selected

accepted formats ▼

List of adapters adapter sequences which will be explicitly searched against the library. It should be a tab-delimited file with 2 columns: name and sequence. (--adapters)

Submodule and Limit specifying file - optional

📄 📄 📁 ⋮ Nothing selected

accepted formats ▼

a file that specifies which submodules are to be executed (default=all) and also specifies the thresholds for the each submodules warning parameter

Disable grouping of bases for reads >50bp

☐ No

Using this option will cause fastqc to crash and burn if you use it on really long reads, and your plots may end up a ridiculous size. You have been warned! (--nogroup)

Lower limit on the length of the sequence to be shown in the report - optional

As long as you set this to a value greater or equal to your longest read length then this will be the sequence length used to create your read groups. This can be useful for making directly comparable statistics from datasets with somewhat variable read lengths. (--min_length)

Length of Kmer to look for *

Note: the Kmer test is disabled and needs to be enabled using a custom Submodule and limits file (--kmers)

Additional Options

defaultni
parametri su
najčešće
dovoljni za
inicijalnu
procjenu

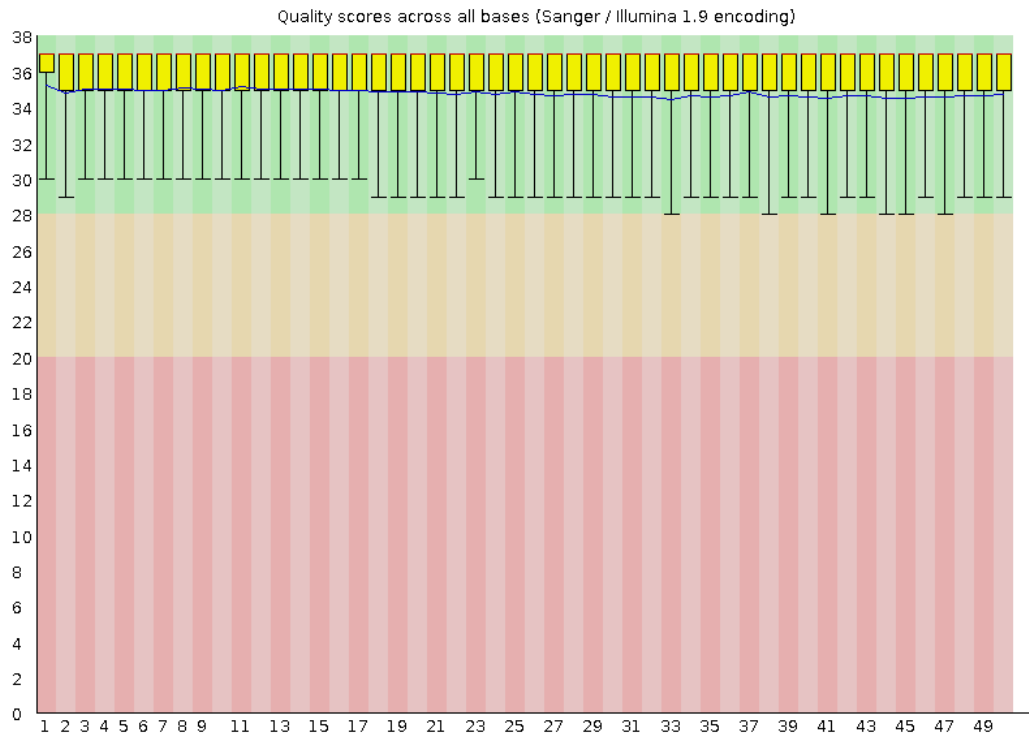
FastQC

- alat za procjenu kvalitete sirovih FASTQ datoteka
- provjerava:
 1. **kvalitetu baza** - *Phred score* po poziciji
 2. **GC udio** - otkriva moguće kontaminacije ili pristranosti
 3. **razinu duplikacije** - ukazuje na PCR preamplifikaciju ili nisku složenost
 4. **prisutnost adaptera** - pokazuje treba li skratiti očitavanja (*trimming*)
 5. **prezastupljenost sekvenci** - moguće kontaminacije ili tehnički artefakti
- provodi se nakon sekvenciranja te ponovno nakon pročišćavanja i kraćenja sekvenci
- [documentation](#)

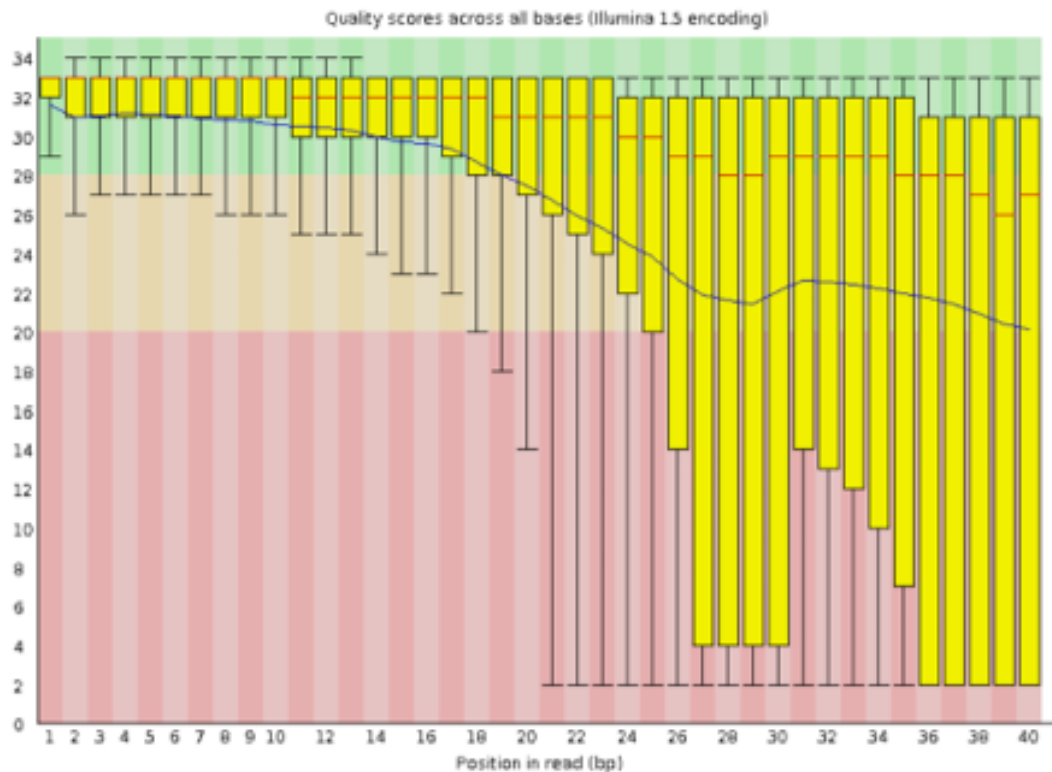
FastQC output: per base sequence quality



Per base sequence quality



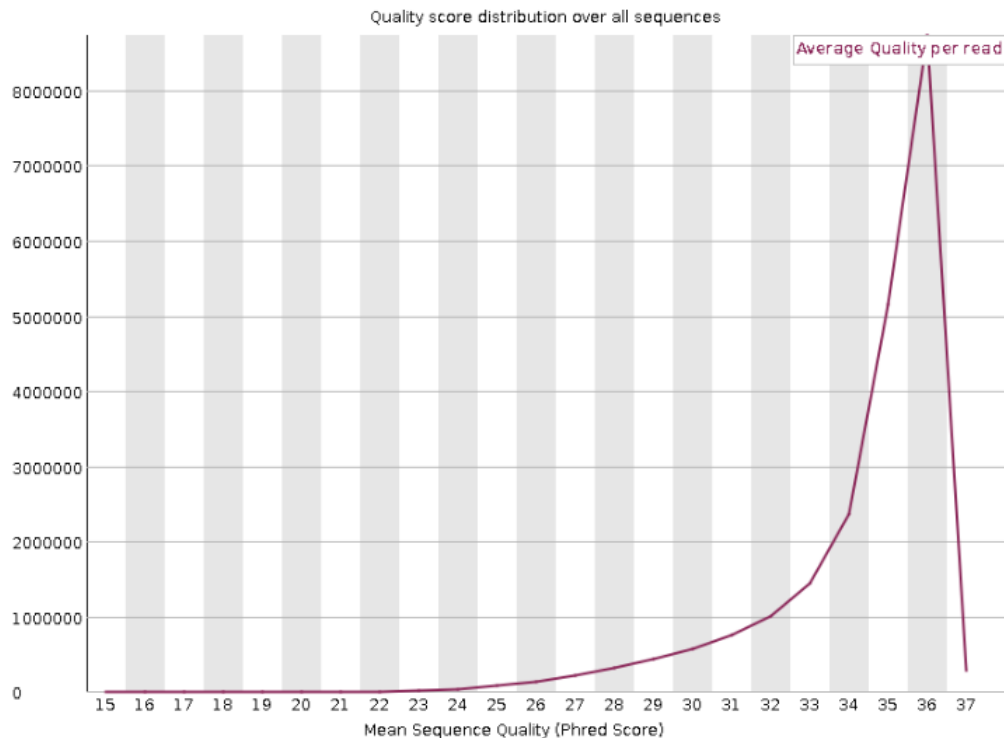
FastQC output: per base sequence quality



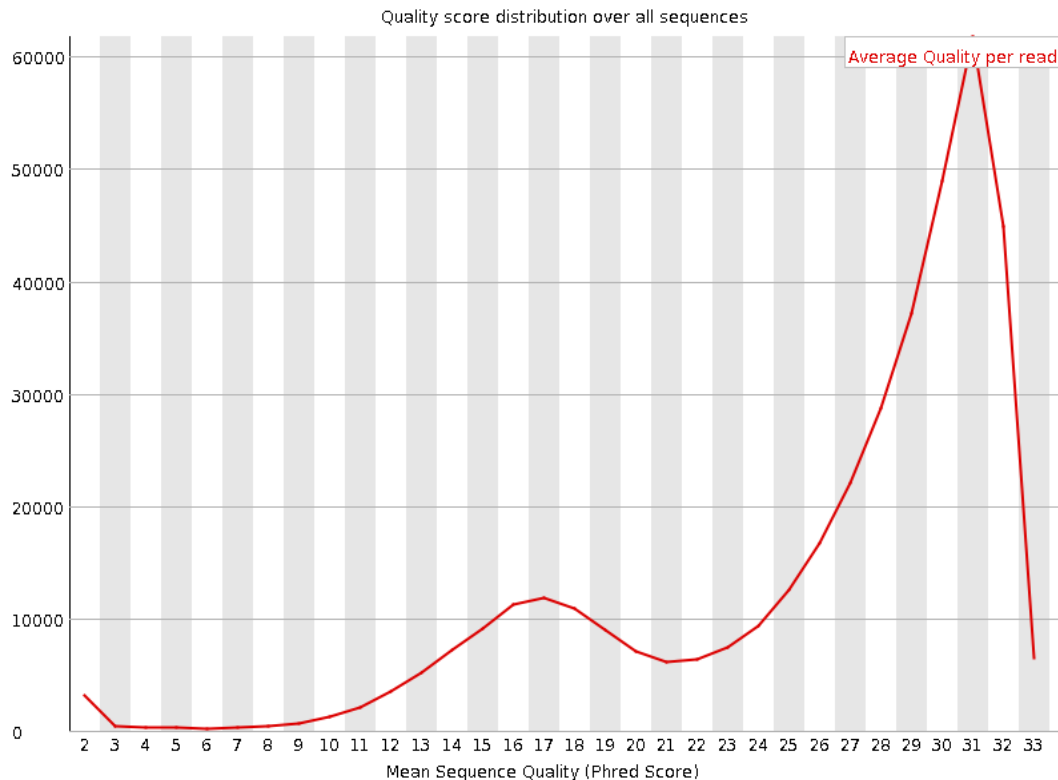
RJEŠENJE:
uklonite krajeve s
niskom kvalitetom
(*trimming*)

FastQC output: per sequence quality scores

✓ Per sequence quality scores



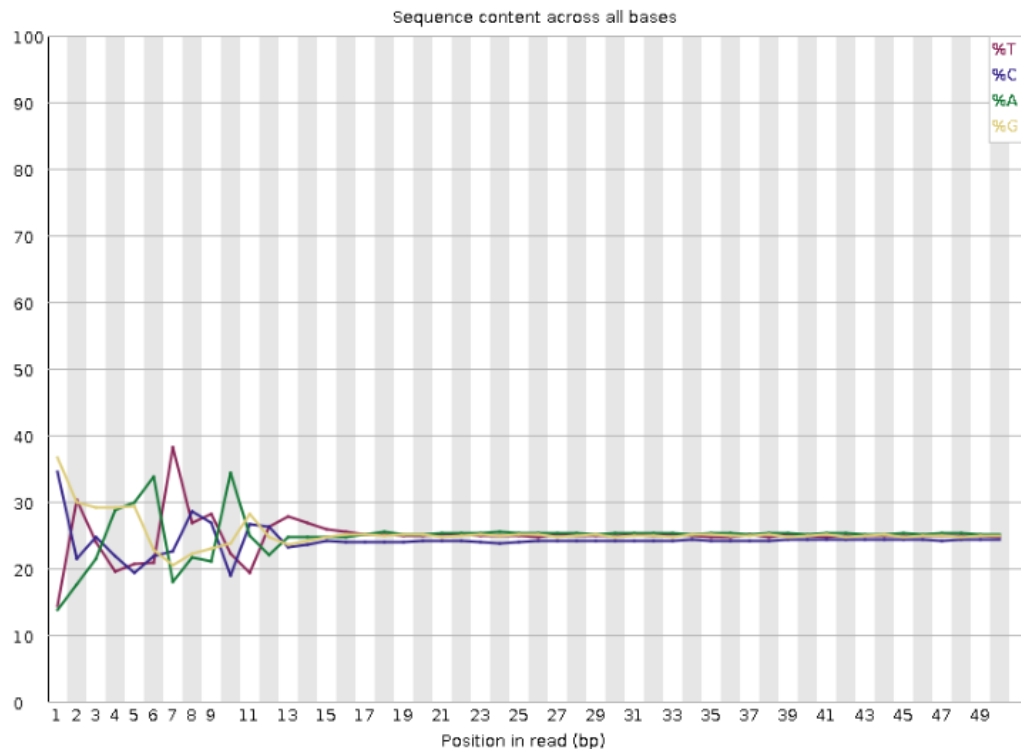
FastQC output: per sequence quality scores



RJEŠENJE:
izbacite readove s
prosječno lošom
kvalitetom (*filtering*)

FastQC output: per base sequence content

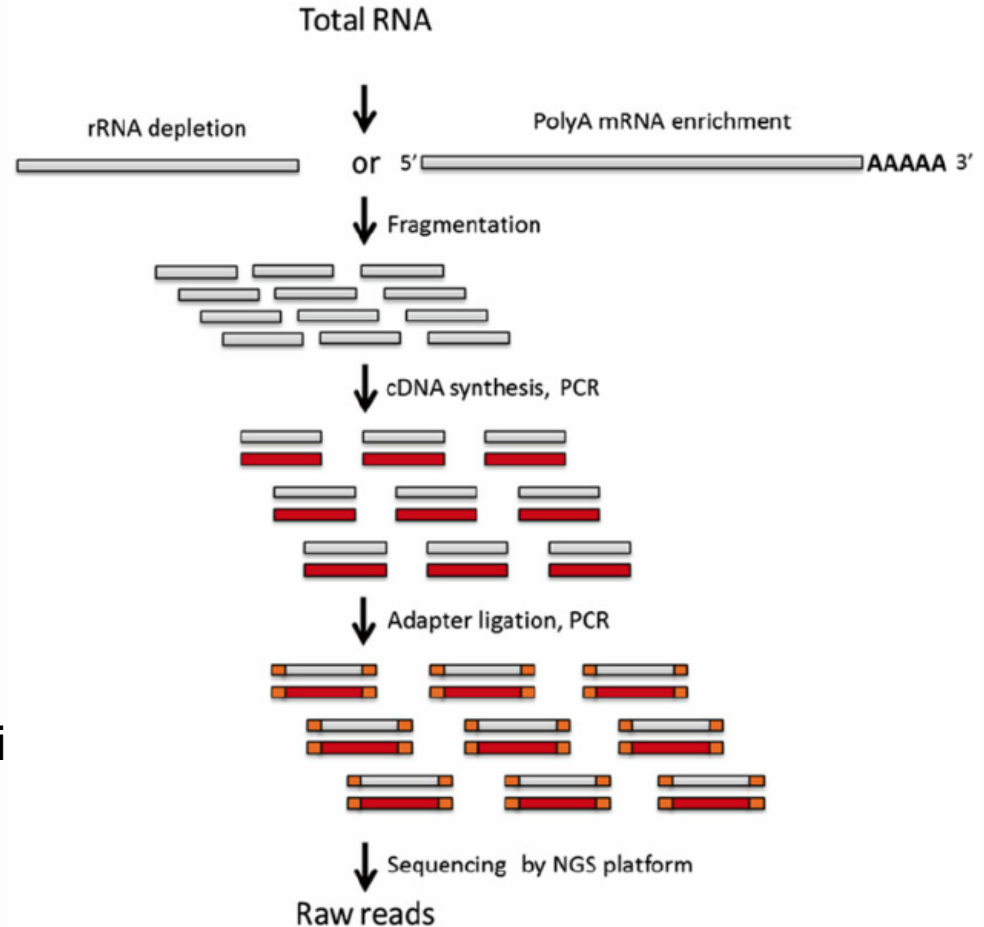
✖ Per base sequence content



RNA-Seq

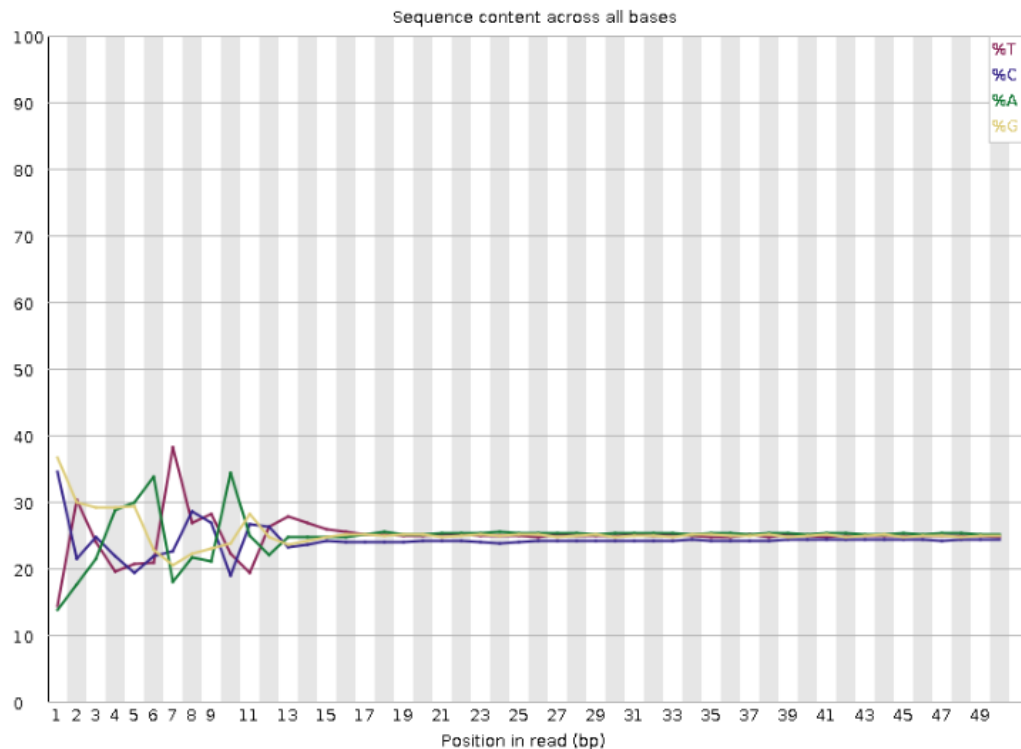
- nespecifične probe
 - "hvataju" sve
 - niža cijena od specifičnih proba
- brojne varijante
- moderni standard

slučajni heksameri

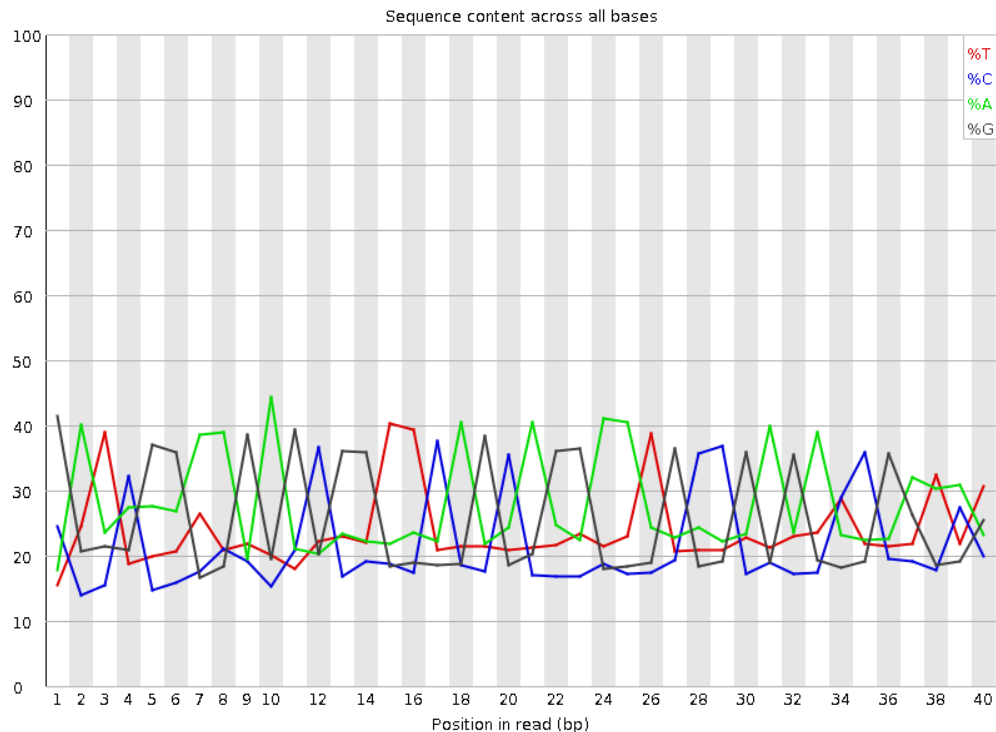


FastQC output: per base sequence content

✖ Per base sequence content



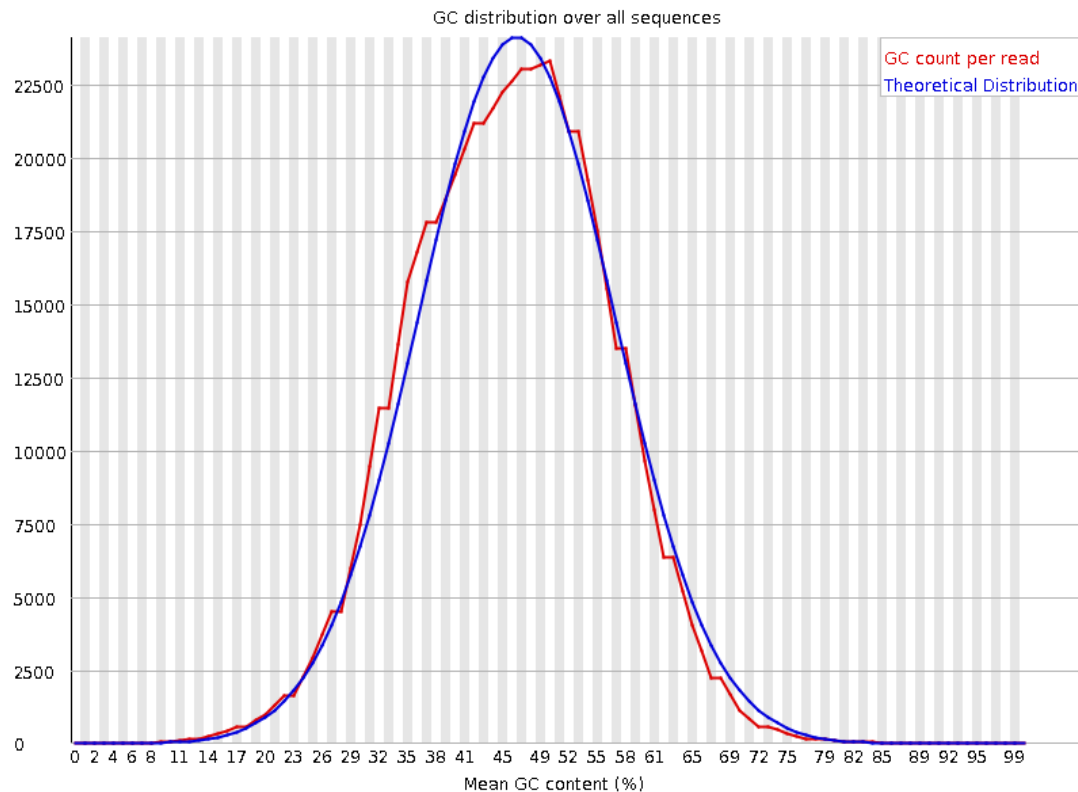
FastQC output: per base sequence content



Neke sekvence su prisutne u neproporcionalno većim količinama (*overrepresented*).

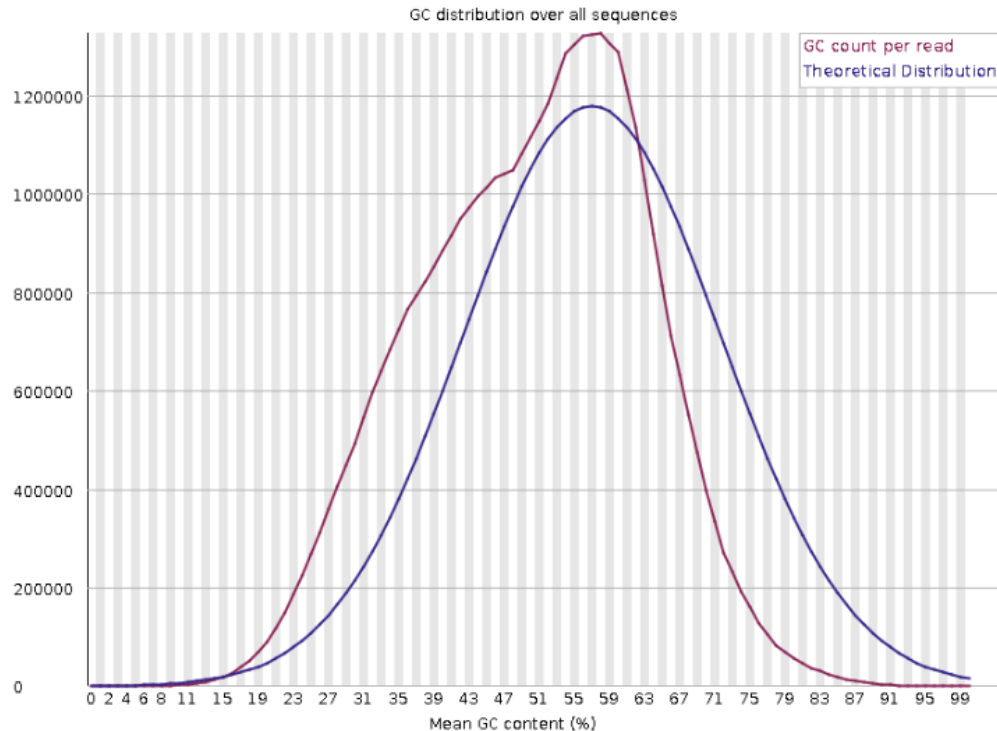
Nije uvijek problem kojeg treba popravljati!

FastQC output: per sequence GC content

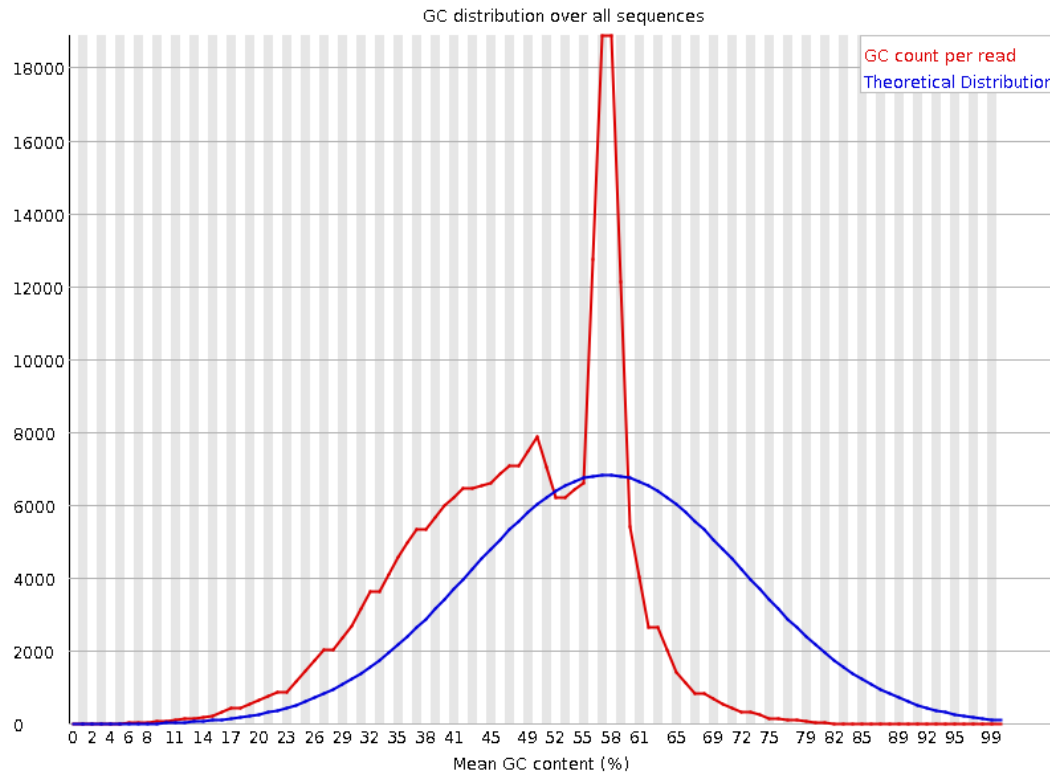


FastQC output: per sequence GC content

✖ Per sequence GC content



FastQC output: per sequence GC content



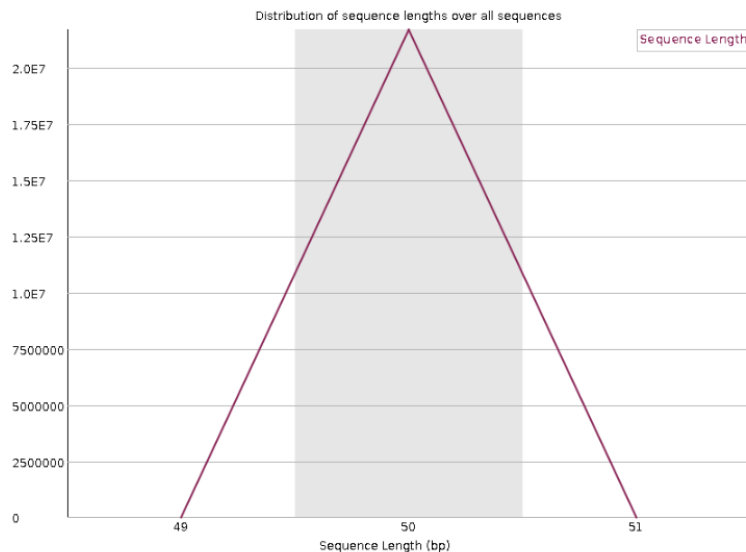
Može biti znak kontaminacije – ili bakterijske ili zbog drugih *overrepresented* sekvenci.

RJEŠENJE:

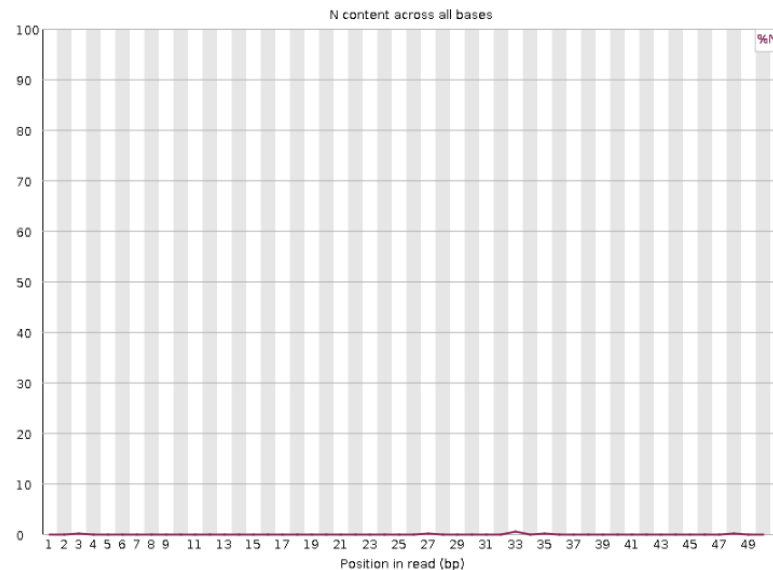
- ne učiniti ništa
- pokušati filtrirati
- ponovno sekvenciranje

FastQC output: sequence length distribution & per-base N content

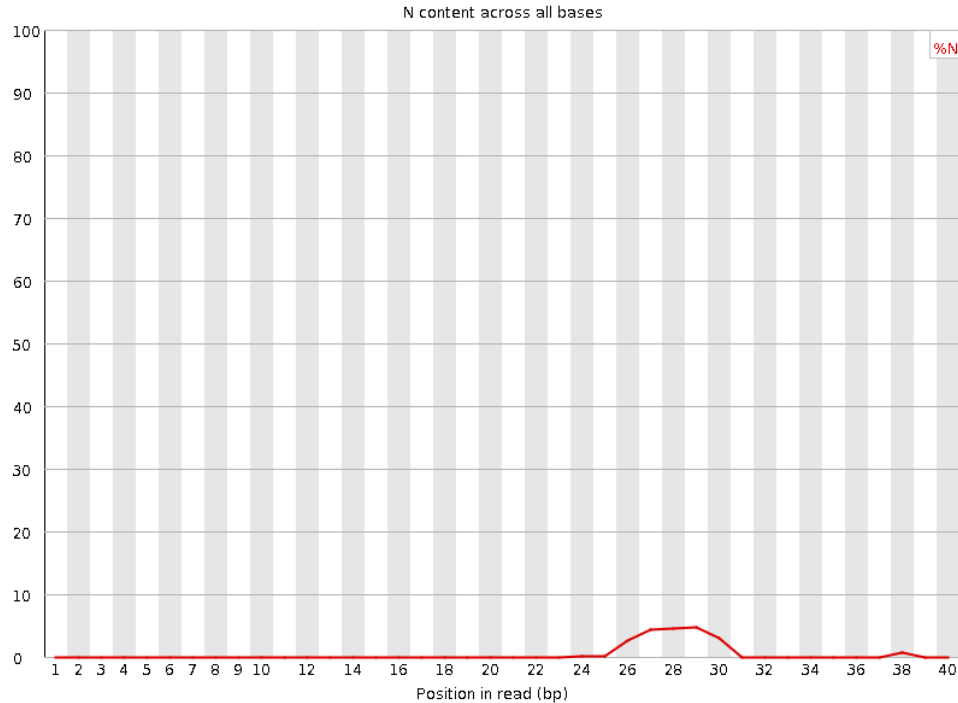
✓ Sequence Length Distribution



✓ Per base N content



FastQC output: per-base N content



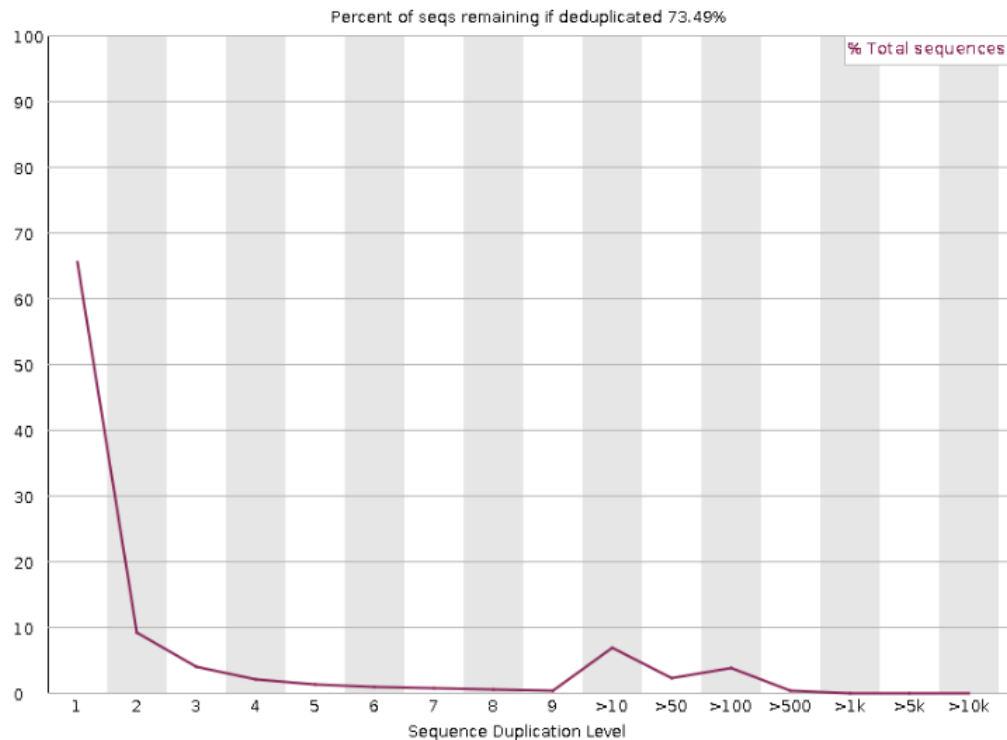
Problem u reakciji
sekvenciranja.

RJEŠENJE: ponoviti
sekvenciranje.

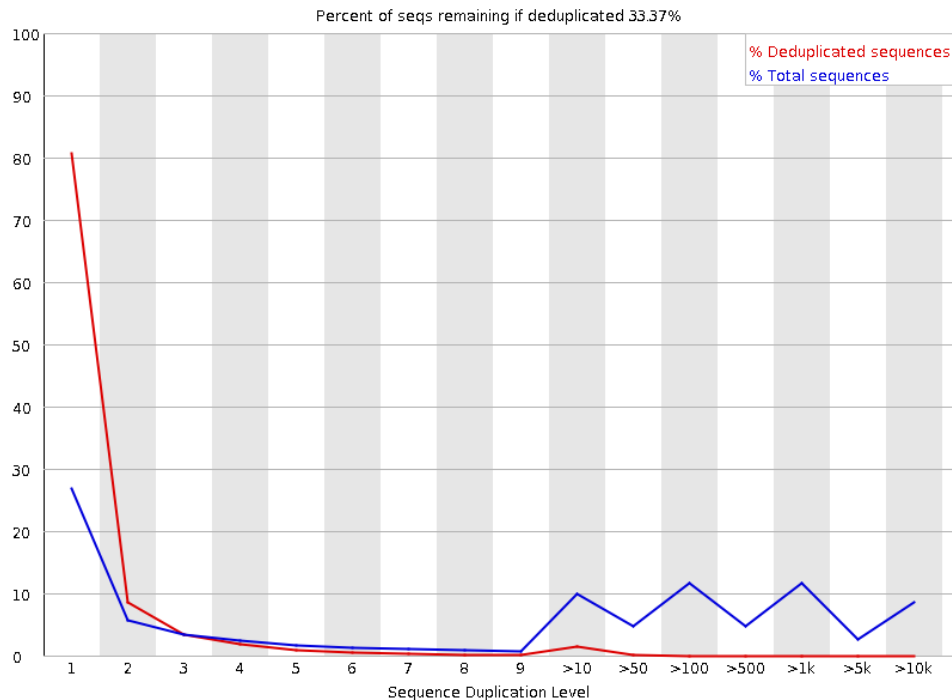
FastQC output: sequence duplication levels



Sequence Duplication Levels



FastQC output: sequence duplication levels

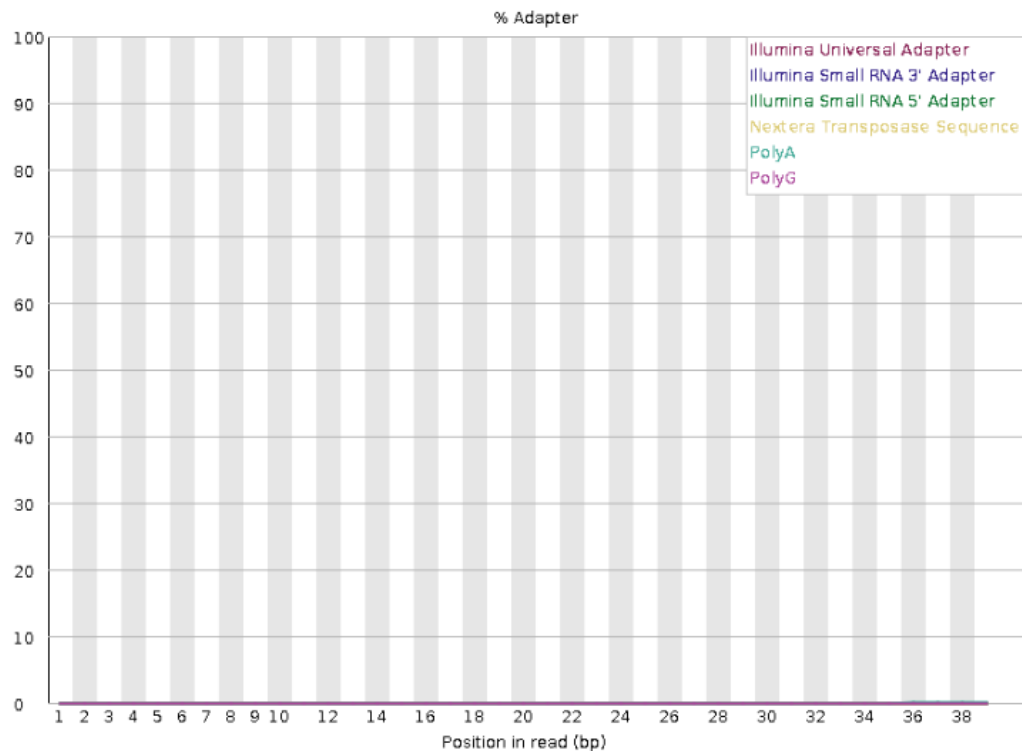


Tehnološki duplikati u
sekvencama, npr.
PCR duplicates.

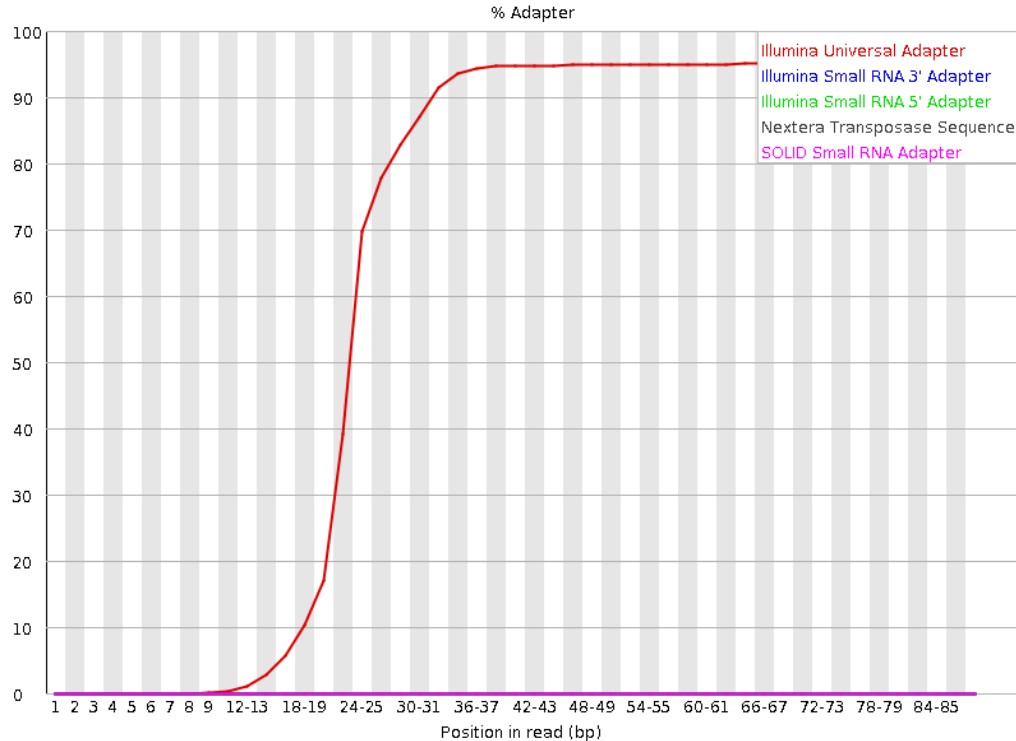
RJEŠENJE:
deduplikacija *readova*
(nakon mapiranja).

FastQC output: adapter content

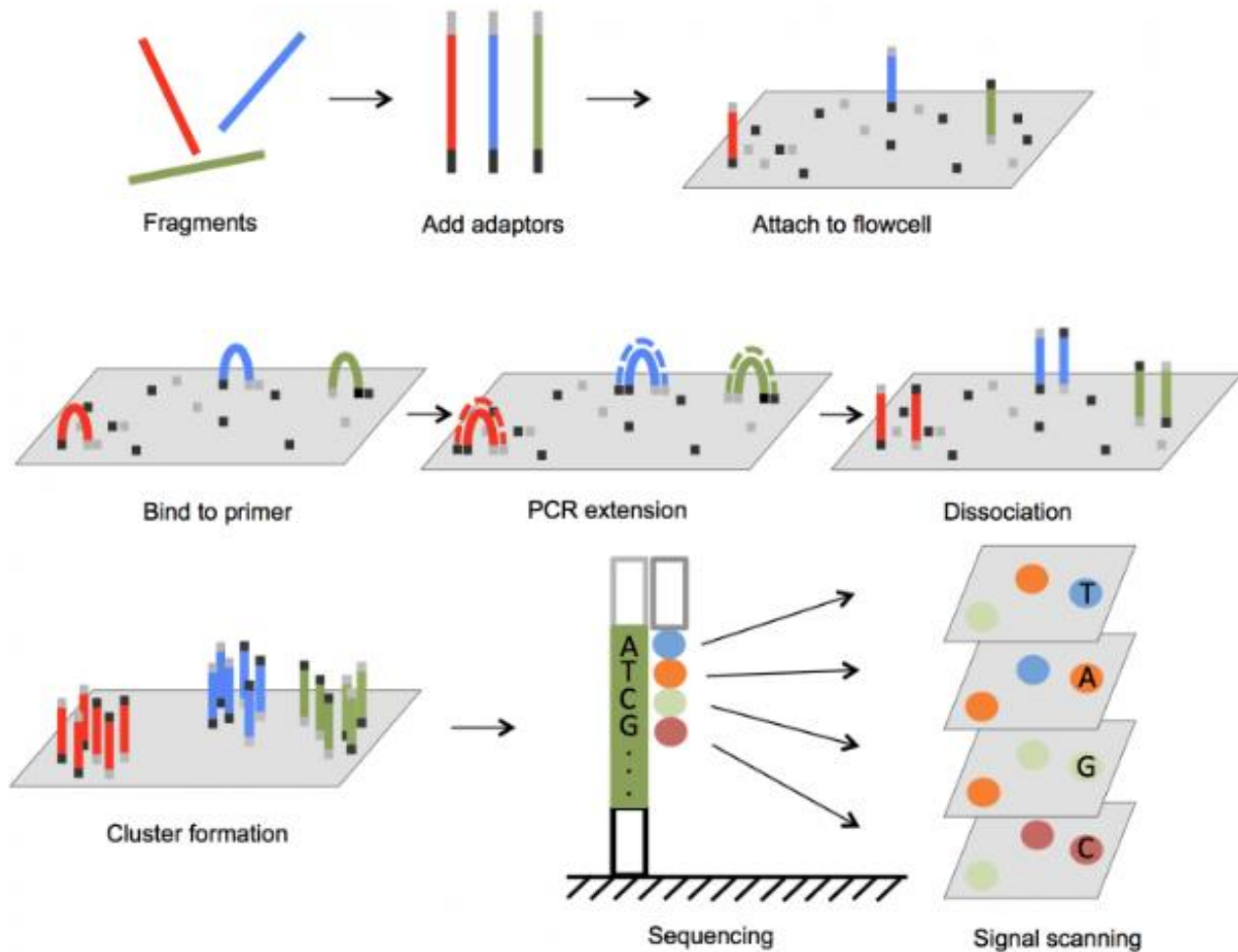
✓ Adapter Content



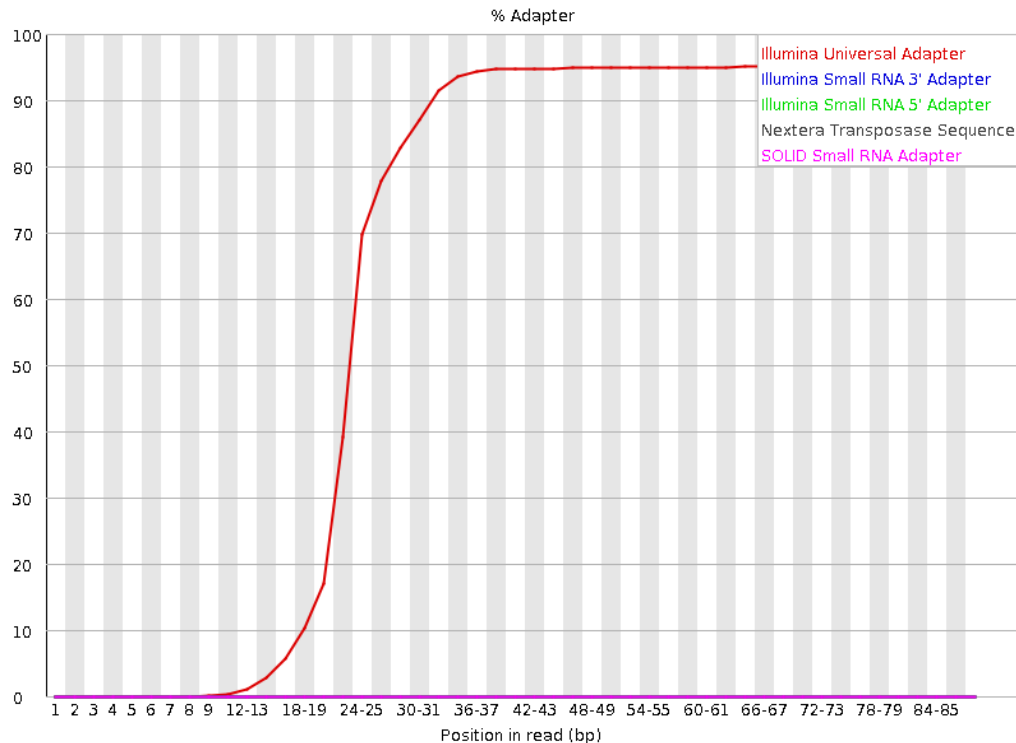
FastQC output: adapter content



Read-through –
adapteri očitani
tijekom sekvenciranja.



FastQC output: adapter content



Read-through –
adapteri očitani
tijekom sekvenciranja.

RJEŠENJE: uklonite
dijelove koji sadrže
adaptere (trimming).

MultiQC

Galaxy Europe



Upload



Tools



ChatGXY



Workflows



Workflow Invocations



Interactive Tools



Visualization



Histories



History Multiview



Download

Tools

Discover Tools

multiqc

Show Sections

MultiQC aggregate results from bioinformatics analyses into a single report

MultiQC aggregate results from bioinformatics analyses into a single report (Galaxy Version 1.33+galaxy3)



Tool Parameters

Results

1: Results

Which tool was used generate logs?

FastQC

Software name

FastQC output

1: FastQC output

Type of FastQC output? *

Raw data

FastQC output - optional



accepted formats

11: FastQC on collection 3: RawData *

switch to column select

+ Insert FastQC output

+ Insert Results

odabir parametara

Preview

Visualize

Details

Edit



v1.33



General Stats

FastQC

Sequence Counts

Sequence Quality Histograms

Per Sequence Quality Scores

Per Base Sequence Content

Per Sequence GC Content

Per Base N Content

Sequence Length Distribution

Sequence Duplication Levels

Overrepresented sequences by sample

Adapter Content

Status Checks

Software Versions



A modular tool to aggregate results from bioinformatics analyses across many samples into a single report.

Report generated on 2026-04-15, 22:34 CEST based on data in: /data/jwd06/main/000/100/100/100100736/working/multiqc_wdir

[+ Summarize report](#)

📘 Welcome! Not sure where to start?

[Watch a tutorial video](#)

(6:06)

[don't show again](#)

General Statistics

[Copy table](#)[Configure columns](#)[Scatter plot](#)[Violin plot](#)[Export as CSV...](#)

Showing 6/6 rows and 3/6 columns.

[+ Summarize table](#)

Sample Name	Dups	GC	Seqs
SRR23318468	26.5 %	49.0 %	21.7 M
SRR23318469	25.3 %	50.0 %	21.7 M
SRR23318470	39.7 %	48.0 %	21.7 M
SRR23318471	29.5 %	50.0 %	21.6 M
SRR23318472	29.1 %	48.0 %	21.6 M
SRR23318473	39.7 %	49.0 %	21.7 M

FastQC Version: 0.12.1

Quality control tool for high throughput sequencing data. <http://www.bioinformatics.babraham.ac.uk/projects/fastqc>

Sequence Counts

[Help](#)

Sequence counts for each sample. Duplicate read counts are an estimate only.

[Percentages](#)[+ Summarize plot](#)[Export...](#)

FastQC: Sequence Counts

6 bars

SRR23318468



Unique Reads

Status Checks

[? Help](#)

Status for each FastQC section showing whether results seem entirely normal (green), slightly abnormal (orange) or very unusual (red).

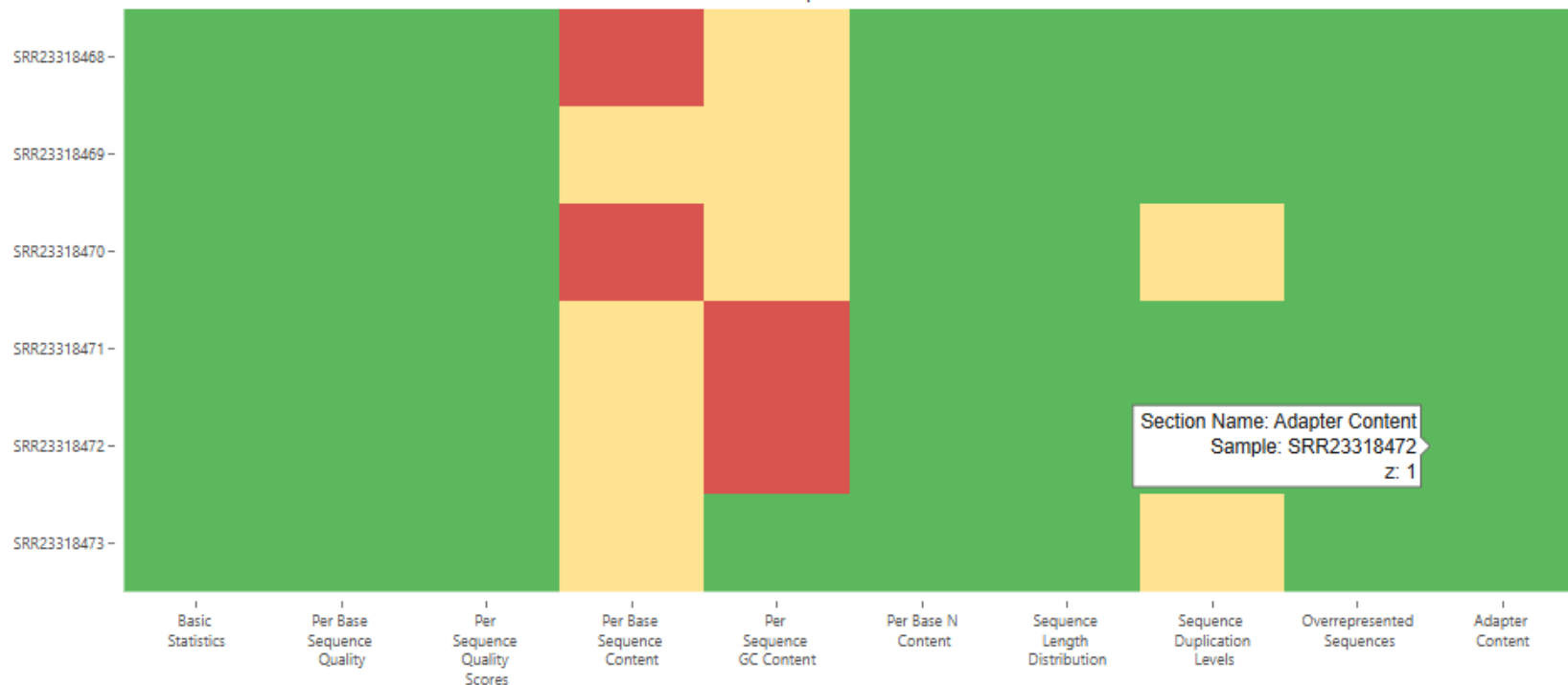
Sorted by sample

Clustered

+ Summarize plot

± Export...

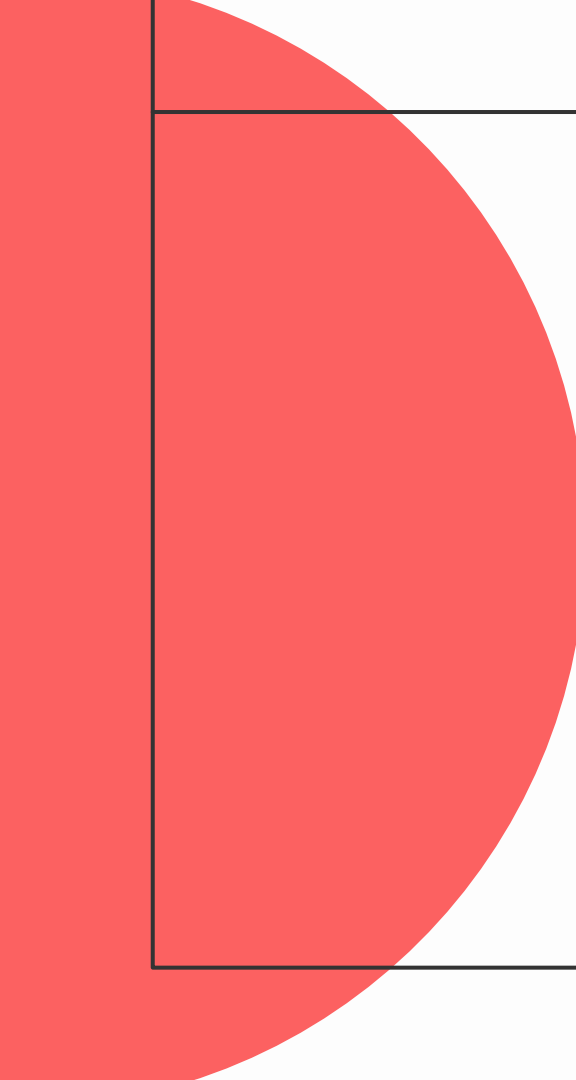
FastQC: Status Checks
6 samples



Section Name: Adapter Content
Sample: SRR23318472
z: 1

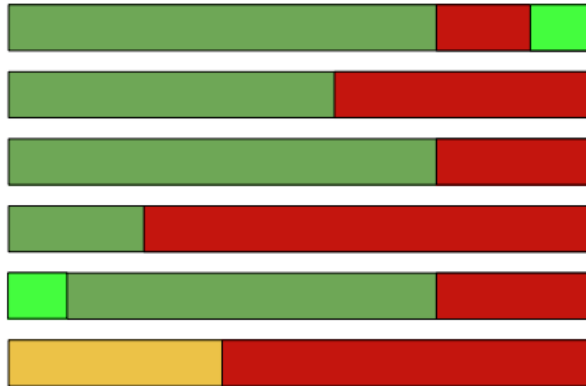
Osnovni koraci

1. QC – kontrola sirovih podataka
2. **kraćenje i pročišćavanje očitavanja** (*trimming and filtering*)
3. ponovna provjera kvalitete
4. mapiranje na genom
5. alignment QC – kontrola mapiranja
6. feature counts
7. normalizacija i statistička analiza

A large red circular graphic element is positioned on the left side of the slide, partially overlapping the white content area.

2. Trimming, filtering i ponovna kontrola kvalitete

Removing adapters



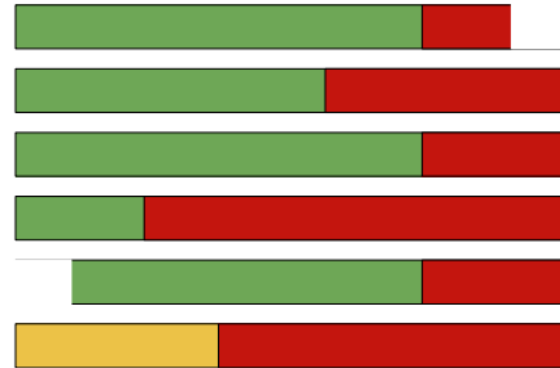
adapter

great

medium

very low

Quality trimming



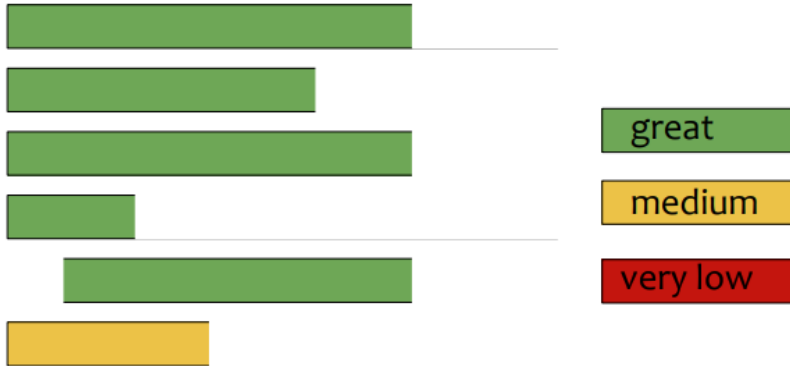
adapter

great

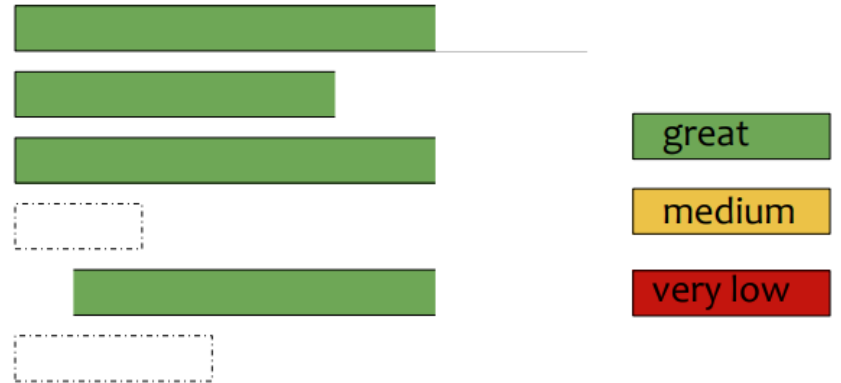
medium

very low

Quality trimming -> quality and size filtering



Quality & size filtering



Trimmomatic

1. **uklanjanje adaptera** – prepozna i uklanja adaptere
2. **skraćivanje na temelju kvalitete** – reže baze s početka ili kraja očitavanja koje imaju nisku kvalitetu, ili koristi “klizni prozor” (*sliding window*) da ukloni dijelove gdje kvaliteta naglo padne
3. **filtriranje na temelju duljine** – uklanja očitavanja koja postanu prekratka nakon skraćivanja
4. **uklanjanje uparenih očitavanja (*paired-end*)** – ukoliko je očitavanje uklonjeno radi loše kvalitete ili kratkoće, uklanja i drugo očitavanje iz para

Trimmomatic

Galaxy Europe

Tools

Discover Tools

trimmo

Show Sections

Trimmomatic flexible read trimming tool for Illumina NGS data

Trimmomatic flexible read trimming tool for Illumina NGS data (Galaxy Version 0.39+galaxy2)

Tool Parameters

Single-end or paired-end reads?

Single-end

Input FASTQ file *

3: Single-end data (fastq-dump)

accepted formats ▾

! The supplied input will be mapped over this tool.

Perform initial ILLUMINACLIP step?

no

Cut adapter and other illumina-specific sequences from the read

Trimmomatic Operation

1: Trimmomatic Operation

Select Trimmomatic operation to perform

Sliding window trimming (SLIDINGWINDOW)

Number of bases to average across *

4

Average quality required *

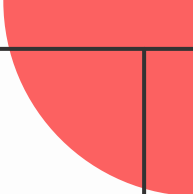
20

+ Insert Trimmomatic Operation

Quality score encoding - optional

Nothing selected

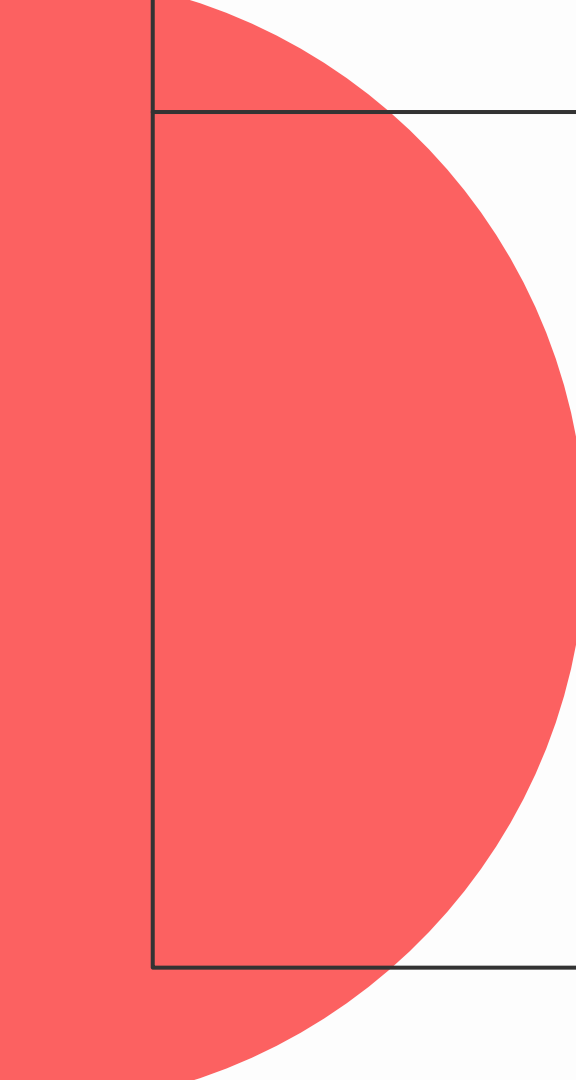
The phred+64 encoding works the same as the phred+33 encoding, except you add 64 to the phred score to determine the ascii code of the quality character. You will only find phred+64 encoding on older data, which was sequenced several years ago. FASTQC can be used in order to identify the encoding type.



Ne zaboravite ponovno provjeriti kvalitetu!

Osnovni koraci

1. QC – kontrola sirovih podataka
2. kraćenje i pročišćavanje očitavanja (*trimming and filtering*)
3. ponovna provjera kvalitete
4. **mapiranje na genom**
5. alignment QC – kontrola mapiranja
6. feature counts
7. normalizacija i statistička analiza

A large red shape, consisting of a semi-circle and a rectangle, is positioned on the left side of the slide.

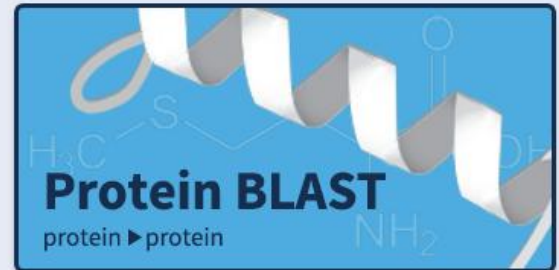
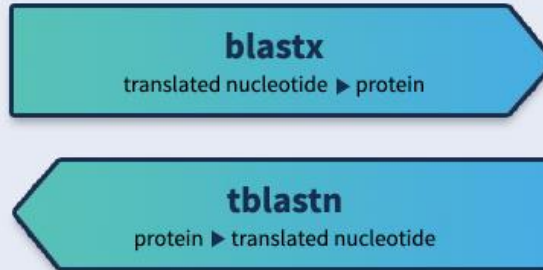
4. Mapiranje na genom

Mapiranje

= proces kojim pokušavamo (jednoznačno) odrediti lokaciju u genomu s koje je potekla očitana sekvenca

BLAST

- Basic Local Alignment Search Tool
- jednostavno korisničko sučelje



blastn

blastp

blastx

tblastn

tblastx

Standard Nucleotide BLAST

BLASTN programs search nucleotide databases using a nucleotide query, more...

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) [Clear](#)

```
GGAAGCAACTCAGAGTCGCCGCACTCAACTAATTCATCCACGCTGCCCT
GAGCCCAGAACTCAAAGGAA
GGATCGGCCCGGGTAGGAGGCCACAGTGATGTCCCCAAAAGTGAATTCT
TCAAAGCCCTTGAGGGC
```

Query subrange [?](#)From To

Or, upload file

No file chosen [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)☐ Align two or more sequences [?](#)

Choose Search Set

Database

☒ Standard databases (nr etc.): ☐ rRNA/ITS databases ☐ Genomic + transcript databases ☐ Betacoronavirus ☐ Experimental databases [?](#)

Organism

Optional

☐

exclude

[Add Organism](#)Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown [?](#)

Exclude

Optional

☐ Models (XM/XP) ☐ Uncultured/environmental sample sequences

Limit to

Optional

☐ Sequences from type material

Entrez® Query

Optional

[Create custom database](#)Enter an Entrez query to limit search [?](#)

Program Selection

Optimize for

- ☒ Highly similar sequences (megablast)
☐ More dissimilar sequences (discontiguous megablast)
☐ Somewhat similar sequences (blastn)

Choose a BLAST algorithm [?](#)

BLAST

Search database **core_nt** using **Megablast** (Optimize for highly similar sequences)☐ Show results in a new window

[← Edit Search](#)[Save Search](#)[Search Summary ▾](#)[? How to read this report?](#)[▶ BLAST Help Videos](#)[↩ Back to Traditional Results Page](#)

Job Title

Nucleotide Sequence

RID

[Y1BSCG2P014](#) Search expires on 04-17 17:42 pm [Download All ▾](#)

Program

BLASTN [?](#) [Citation ▾](#)

Database

core_nt [See details ▾](#)

Query ID

lcl|Query_8235553

Description

None

Molecule type

dna

Query Length

8814

Other reports

[Distance tree of results](#) [MSA viewer](#) [?](#)

Filter Results

Organism only top 20 will appear☐

exclude

[+ Add organism](#)

Percent Identity

E value

Query Coverage

to

to

to

[Filter](#)[Reset](#)

Descriptions

[Graphic Summary](#)[Alignments](#)[Taxonomy](#)

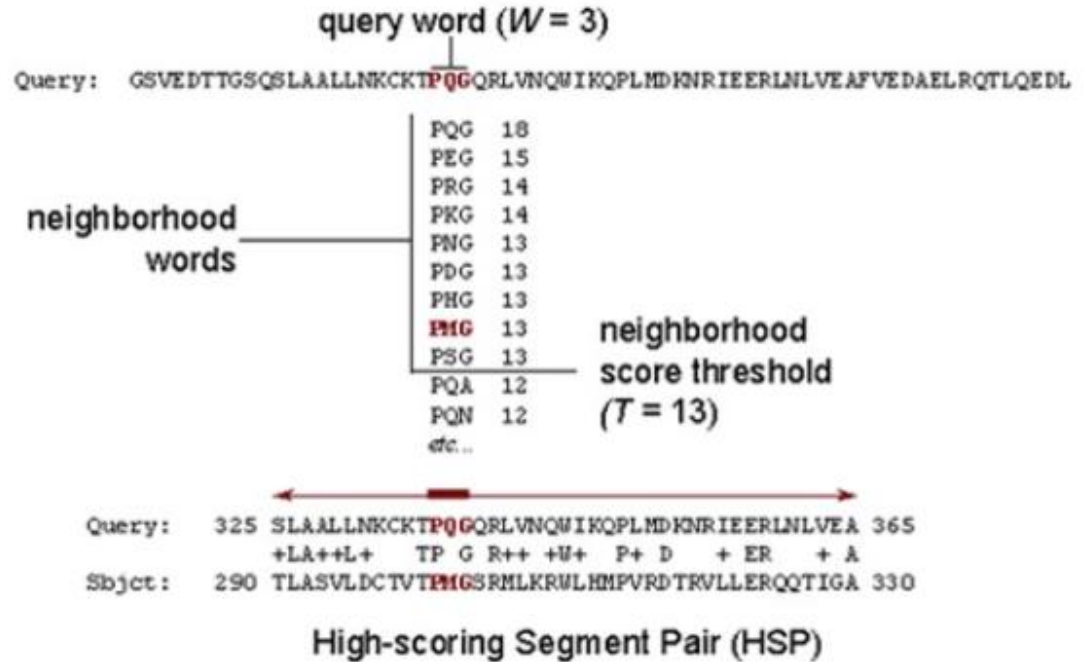
Sequences producing significant alignments

[Download ▾](#)[Select columns ▾](#)[Show](#)[?](#)☒ select all 36 sequences selected[GenBank](#)[Graphics](#)[Distance tree of results](#)[MSA Viewer](#)

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	Homo sapiens chromosome X clone XXcos-LN0617, XXcos-162H10, I104-J0935, XXcos-M1937, XXcos-...	Homo sapiens	16277	16277	100%	0.0	100.00%	167488	U82695.3
☒	Homo sapiens chromosome X clone RP11-54I20, RP11-66N11 map q28, complete sequence	Homo sapiens	16277	16277	100%	0.0	100.00%	191983	AF274858.3
☒	Homo sapiens biglycan (BGN), RefSeqGene on chromosome X	Homo sapiens	16277	16277	100%	0.0	100.00%	21594	NG_015961.1
☒	Pan troglodytes chromosome X clone CH251-572N17 map Xq28, complete sequence	Pan troglodytes	15463	15463	100%	0.0	98.35%	227066	CT737396.3
☒	Homo sapiens HAUS augmin like complex subunit 7 (HAUS7), RefSeqGene on chromosome X	Homo sapiens	9463	9463	58%	0.0	100.00%	54861	NG_012688.3
☒	Homo sapiens chromosome X clone Qc-7G6, QLL-F1720, QLL-C1335, Qc-8B7, Qc-11H12, Qc-7F6, QLL...	Homo sapiens	1489	1489	10%	0.0	98.13%	247592	U52111.3
☒	Pan troglodytes chromosome X clone PTB-029D14 map human ortholog q28, complete sequence	Pan troglodytes	1478	1478	10%	0.0	97.89%	154019	AC151486.1
☒	PREDICTED: Chlorocebus sabaeus HAUS augmin like complex subunit 7 (HAUS7), transcript variant X2...	Chlorocebus sa...	472	472	4%	7e-127	93.40%	1505	XM_007993050.3

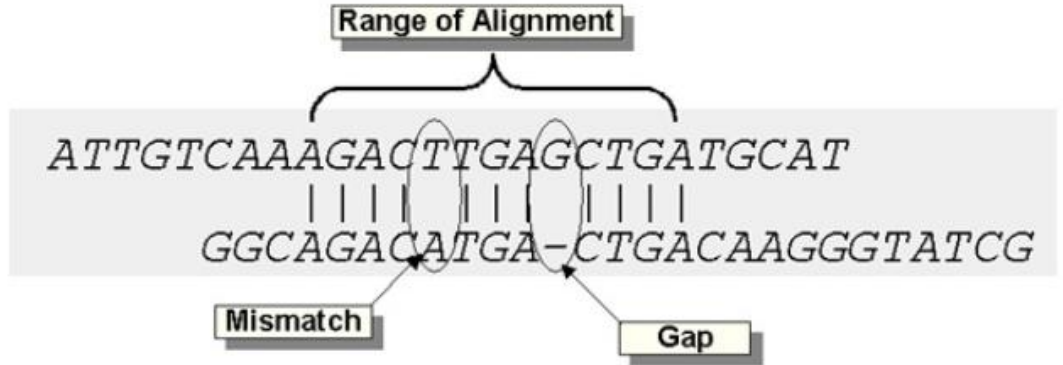
BLAST algoritam

1. podjela na k-mere
2. supstitucijske matrice
3. lista visoko bodovanih riječi



BLAST algoritam

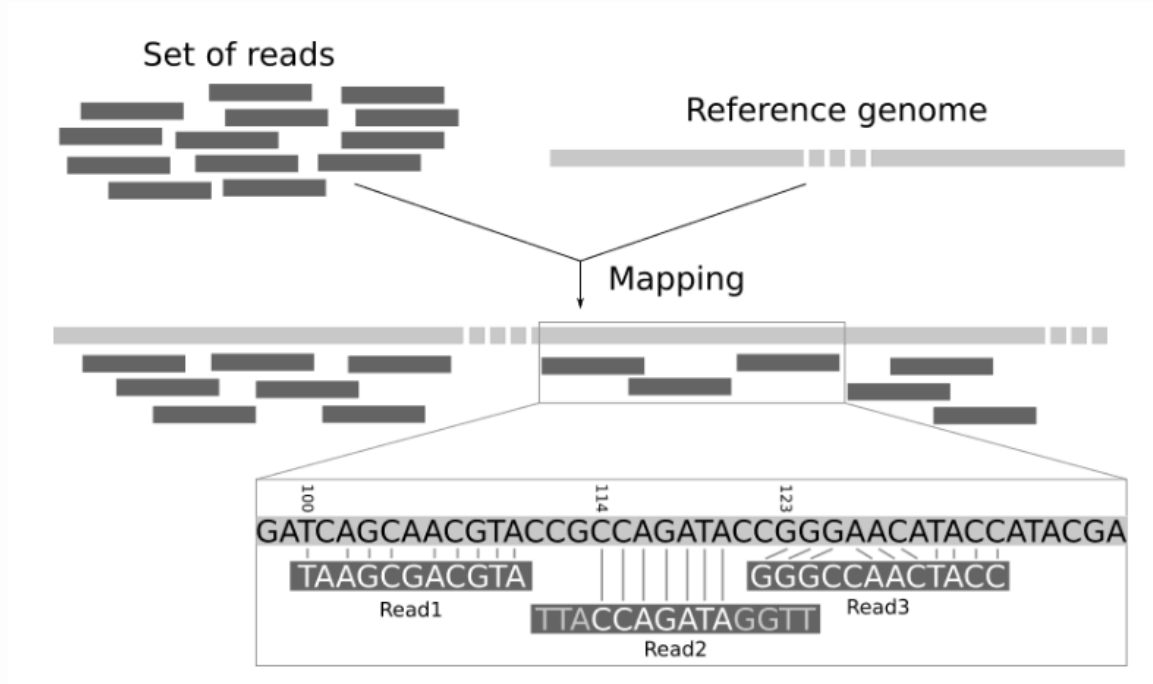
1. podjela na k-mere
2. supstitucijske matrice
3. lista visoko bodovanih riječi
4. točni *matchevi*
5. produljenje sravnjenja



$$S = \sum(\text{identities, mismatches}) - \sum(\text{gap penalties})$$

$$\text{Score} = \text{Max}(S)$$

Mapiranje

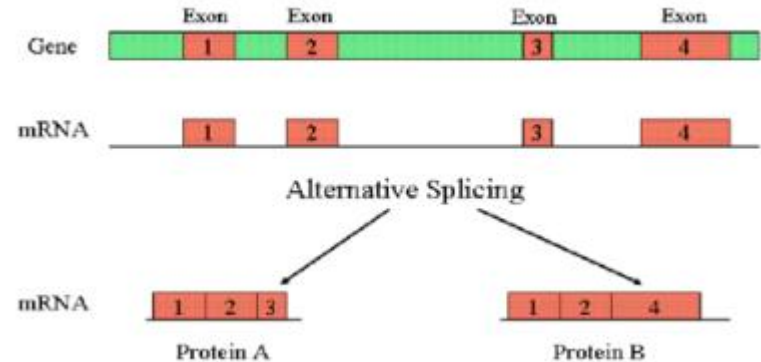
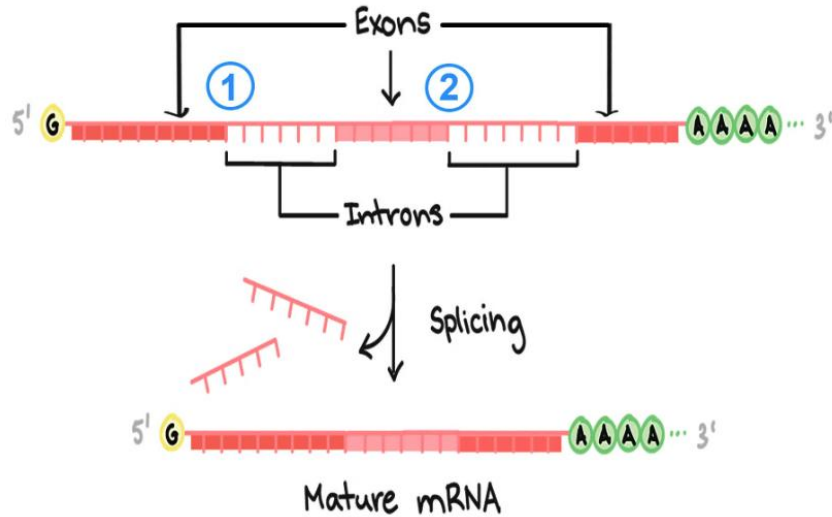


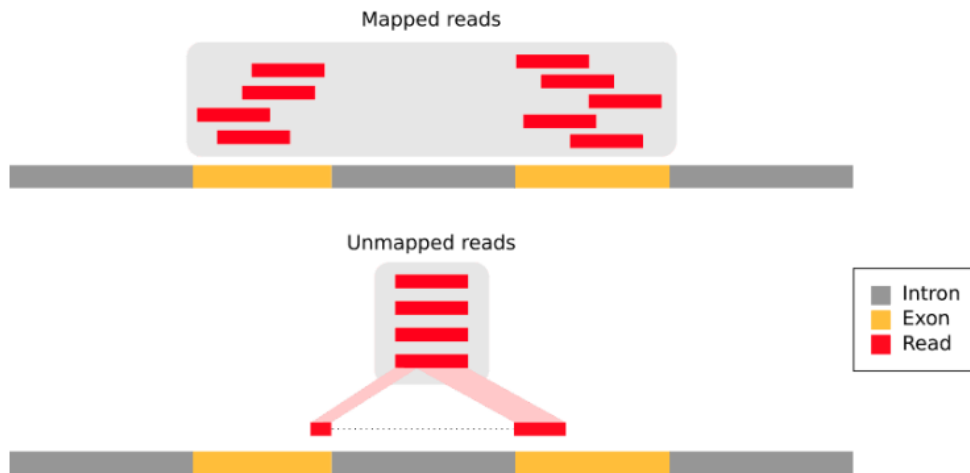
- podjela na k-mere
- točna sravnjenja
- produljenje sravnjenja

Mapiranje

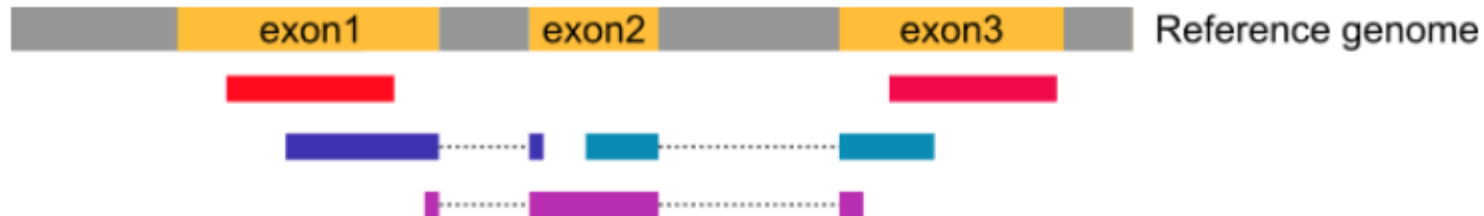
- problem optimalnog smještanja očitane sekvence na genom/referencu
- ponavljajuće sekvence
- prekidi u genomu
- pogreške u sekvenciranju
- PCR duplikati...

Mapiranje RNA - izazov

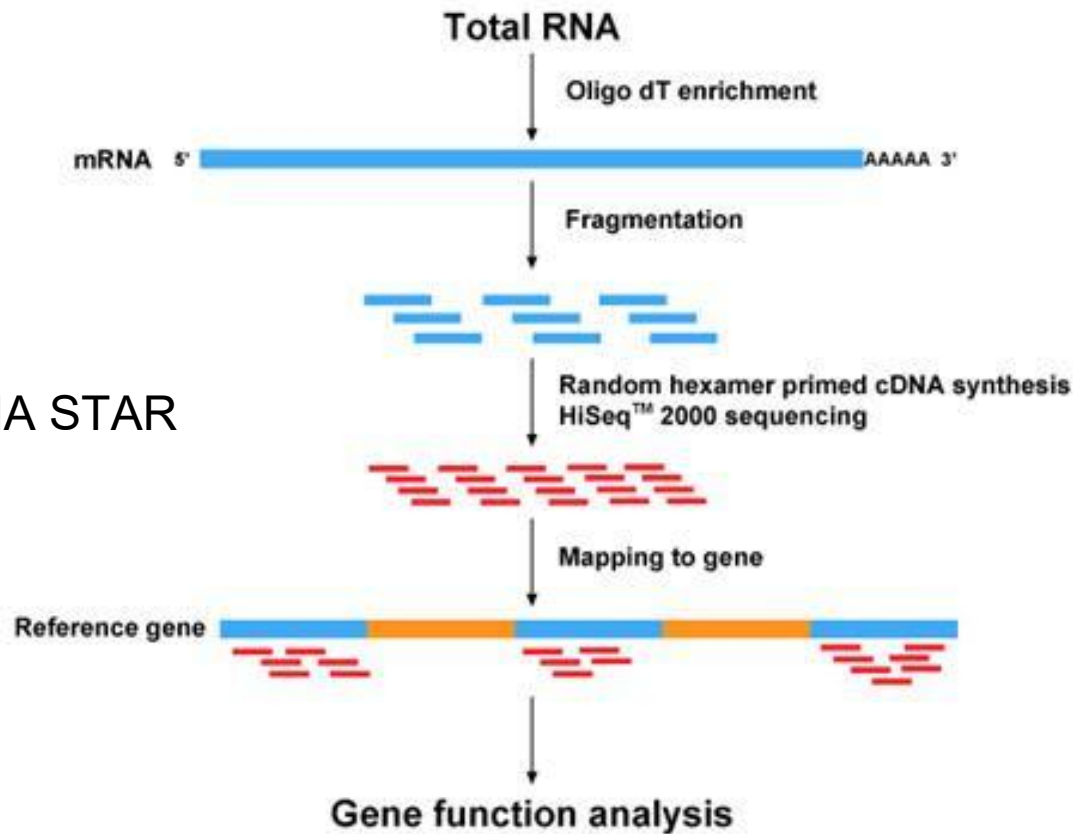




splice-aware mappers

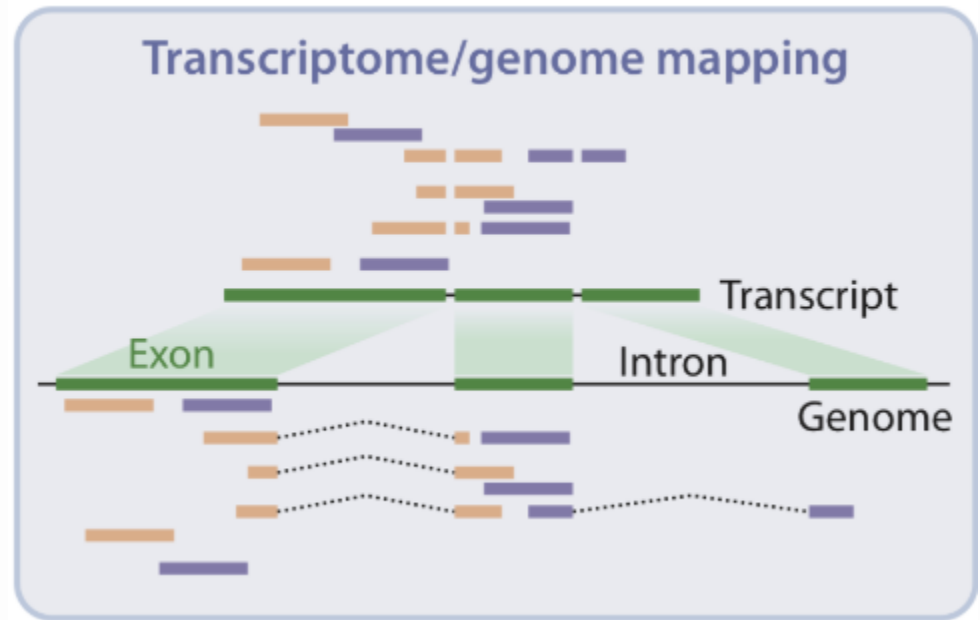


splice-aware mappers:
Bowtie2, minimap2, RNA STAR



Mapiranje RNA-seq podataka

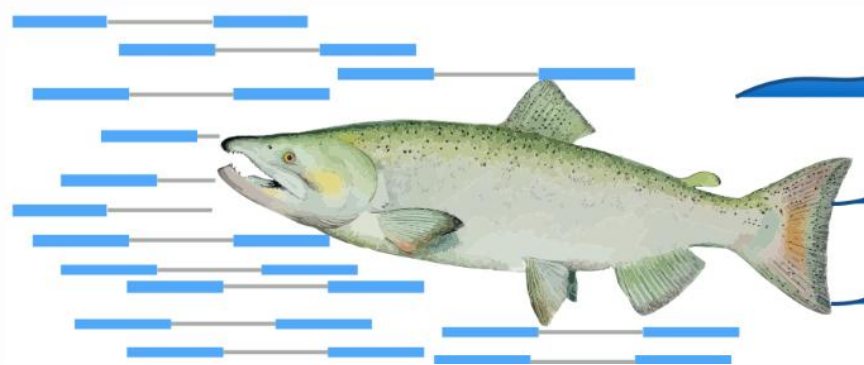
1. na referentni genom
2. na referentni transkriptom



Kvazimapiranje na transkriptom

- *readovi* se ne mapiraju precizno na transkriptom, već se rascijepaju na kraće k-mere
- određuje se vjerojatnost da k-meri pripadaju određenom transkriptu

Salmon, RapMap



RNA STAR

Galaxy Europe



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RNA STAR Gapped-read mapper for RNA-seq data

RNA STARSolo mapping, demultiplexing and gene quantification for single cell RNA-seq

RNA STAR Gapped-read mapper for RNA-seq data (Galaxy Version 2.7.11b+galaxy1)



Tool Parameters

Single-end or paired-end reads

Single-end

RNA-Seq FASTQ/FASTA file *

... 3: Single-end data (fastq-dump)

accepted formats ▼

! The supplied input will be [mapped over](#) this tool.

Custom or built-in reference genome

Use a built-in index

Built-ins were indexed using default options

Reference genome with or without an annotation

use genome reference without builtin gene-model but provide a gtf

Select the '... with builtin gene-model' option to select from the list of available indexes that were built with splice junction information. Select the '... without builtin gene-model' option to select from the list of available indexes without annotated splice junctions, and, optionally, provide your own splice-junction annotations.

Select reference genome *

A. mellifera genome (apiMel3, Baylor HGSC Amel3.0)

If your genome of interest is not listed, contact the Galaxy team (--genomeDir)

Referentna anotacija

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exclude restricted

Name	Description	Synopsis
covid-19	Public data related to COVID-19	RAW sequence data & VCFs
Critical Assessment of Metagenome Interpretation (CAMI)	CAMI datasets	
Earth System Community Modeling	Input data used for running simulation w ... (more)	Earth System Community Modeling data
Galaxy courses	Data for Galaxy courses	
Genomes + annotations	Reference genomes and gene annotations	
GTN - Material	Galaxy Training Network Material	Galaxy Training Network Material. See ht ... (more)
IAEA workshop	workshop data	
Mira_test	for testing	
PGP-UK Open Access Data	Open Access Genomic, Transcriptomic and ... (more)	https://www.personalgenomes.org.uk
Street Science Community	Data for the Street Science Community	

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per page, 10 total

Referentna anotacija

  Add to History  Download  Details

Libraries / Genomes + annotations

	Name	Description	Type	Size	Updated	State
	 Annotations	Gene annotations	folder		over 7 years ago	
	 DTOL	Darwin Tree of Life (https://www.darwint...) (more)	folder		over 2 years ago	
	 Genome in a bottle	NIST's Genome in a Bottle (GIAB) project ... (more)	folder		over 5 years ago	
	 Genomes	Reference genomes	folder		almost 5 years ago	
	 Glimmer Trainingssets	Glimmer Trainingssets	folder		over 12 years ago	
	 iGenomes		folder		over 6 years ago	
	 MSigDB	MSigDB Collections (http://www.broadinst...) (more)	folder		about 12 years ago	
	 RepBase	RepBase RepeatMasker Edition 2017-01-27	folder		over 5 years ago	
	 RNAs	Collection of RNA classes	folder		almost 12 years ago	
	 SARS-CoV-2	Initial analysis of COVID-19 data using ... (more)	folder		about 6 years ago	

Referentna anotacija

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Details

Libraries / Genomes + annotations / Annotations

	Name	Description	Type	Size	Updated	State
Folder	2017		folder		about 9 years ago	
Folder	old		folder		almost 11 years ago	
File	Danio_rerio.GRCz10.91.gtf (danRer10)	Warning: The file 'Type' was set to 'gt ... (more)	gtf	440.3 MB	over 7 years ago	Public
File	Drosophila_melanogaster.BDGP5.78.gtf (dm3)	uploaded gtf file	gtf	152 MB	about 9 years ago	Public
File	Drosophila_melanogaster.BDGP6.87.gtf (dm6)	uploaded gtf file	gtf	148.8 MB	about 9 years ago	Public
File	Homo_sapiens.GRCh37.75.gtf (hg19)	uploaded gtf file	gtf	816.7 MB	about 9 years ago	Public
File	Homo_sapiens.GRCh38.87.gtf (hg38)	uploaded gtf file	gtf	1.3 GB	about 9 years ago	Public
File	Mus_musculus.GRCm38.87.gtf (mm10)	uploaded gtf file	gtf	872.8 MB	about 9 years ago	Public
File	Mus_musculus.NCBIIM37.67.gtf (mm9)	uploaded gtf file	gtf	277.8 MB	about 9 years ago	Public
File	P.patens.V1.6.gtf	uploaded gtf file	gtf	45.1 MB	over 11 years ago	Public

«

«

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>

»

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per page, 12 total

Upload



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ChatGXY



Workflows



Workflow

Invocations



Interactive



Visualization



Histories



History

Multiview



Datasets



Pages



Libraries



Recent

Exports

Libraries / Genomes + annotations / Annotations / Homo_sapiens.GRCh38.87.gtf (hg38)

Download

to History

This dataset is unrestricted so everybody with the link can access it.

Name	Homo_sapiens.GRCh38.87.gtf (hg38)
Data type	gtf
Genome build	hg38
Size	1.3 GB
Date last updated (UTC)	2017-01-18T10:51:28.803005
Date uploaded (UTC)	2017-01-17T10:57:45.745499
Uploaded by	videmp@informatik.uni-freiburg.de
Number of Columns	9
Column Types	str, str, str, int, int, str, str, str, str
Miscellaneous blurb	~2,700,000 lines
Miscellaneous information	uploaded gtf file
Tags	[]
UUID	c9f3155d-5d3c-4edc-91e9-0ff8fd8d195d

1.Seqname 2.Source 3.Feature 4.Start 5.End 6.Score 7.Strand 8.Frame 9.Attributes

chr1	havana	gene	11869	14409	.	+	.	gene_id "ENSG00000223972"; gene_version "5"; gene_name "DDX11L1"; gene_source "havana"; gene_biotype "transcribed_unprocessed_pseudogene"; havana_gene "OTTHUMG00000000961"; havana_gene_version "2";
chr1	havana	transcript	11869	14409	.	+	.	gene_id "ENSG00000223972"; gene_version "5"; transcript_id "ENST00000456328"; transcript_version "2"; gene_name "DDX11L1"; gene_source "havana"; gene_biotype "transcribed_unprocessed_pseudogene"; havana_gene "OTTHUM
chr1	havana	exon	11869	12227	.	+	.	gene_id "ENSG00000223972"; gene_version "5"; transcript_id "ENST00000456328"; transcript_version "2"; exon_number "1"; gene_name "DDX11L1"; gene_source "havana"; gene_biotype "transcribed_unprocessed_pseudogene"; havana_g
chr1	havana	exon	12613	12721	.	+	.	gene_id "ENSG00000223972"; gene_version "5"; transcript_id "ENST00000456328"; transcript_version "2"; exon_number "2"; gene_name "DDX11L1"; gene_source "havana"; gene_biotype "transcribed_unprocessed_pseudogene"; havana_g
chr1	havana	exon	13221	14409	.	+	.	gene_id "ENSG00000223972"; gene_version "5"; transcript_id "ENST00000456328"; transcript_version "2"; exon_number "3"; gene_name "DDX11L1"; gene_source "havana"; gene_biotype "transcribed_unprocessed_pseudogene"; havana_g

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RNA STAR Gapped-read mapper for RNA-seq data (Galaxy Version 2.7.11b+galaxy1)

Use a built-in index

☆ ⚙️ ▼ 📄 ▶ Run Tool

Built-ins were indexed using default options

Reference genome with or without an annotation

use genome reference without builtin gene-model but provide a gtf

Select the '... with builtin gene-model' option to select from the list of available indexes that were built with splice junction information. Select the '... without builtin gene-model' option to select from the list of available indexes without annotated splice junctions, and, optionally, provide your own splice-junction annotations.

Select reference genome *

Human Dec. 2013 (GRCh38/hg38) (hg38)

If your genome of interest is not listed, contact the Galaxy team (--genomeDir)

Gene model (gff3,gtf) file for splice junctions *

26: Homo_sapiens.GRCh38.87.gtf (hg38)

accepted formats ▼

Exon junction information for mapping splices (--sjdbGTFfile)

Elements to use from the gene model to use for splice junctions - optional

exon

By default and for almost all cases: 'exon', referring to finding junctions at the RNA splice sites. This can optionally be changed to allow splicing at other levels, such as 'gene', 'transcript', 'CDS'. (--sjdbGTFfeatureExon)

Length of the genomic sequence around annotated junctions *

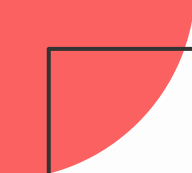
49

base. Ideal value is ReadLength-1 (--sjdbOverhang)

Per gene/transcript output

Per gene read counts (GeneCounts)

STAR can provide analysis results not only with respect to the reference genome, but also with respect to genes and transcripts described by a gene model. Note: This functionality requires either the selection above of a cached index with a gene model, or a gene model provided alongside the index/reference genome in GTF or GFF3 format! (--quantMode)



Take-home message

1. **QC! QC! QC!**
2. standardna analiza, mali broj uzoraka? Galaxy!
3. nestandardna analiza ili puno uzoraka? tražite
(savjet) bioinformatičara